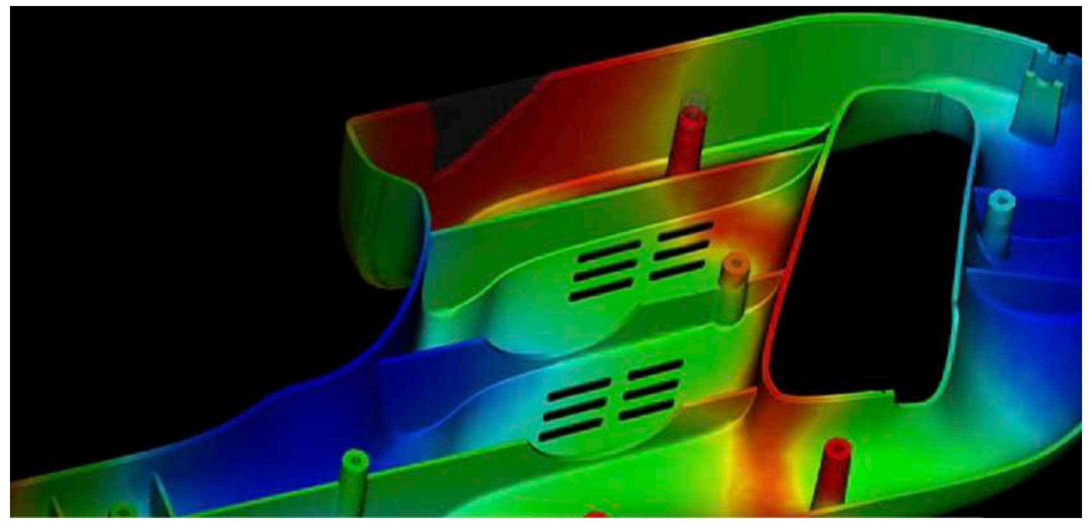
# AI for Design and Manufacturing
## Part 1: Manufacturing

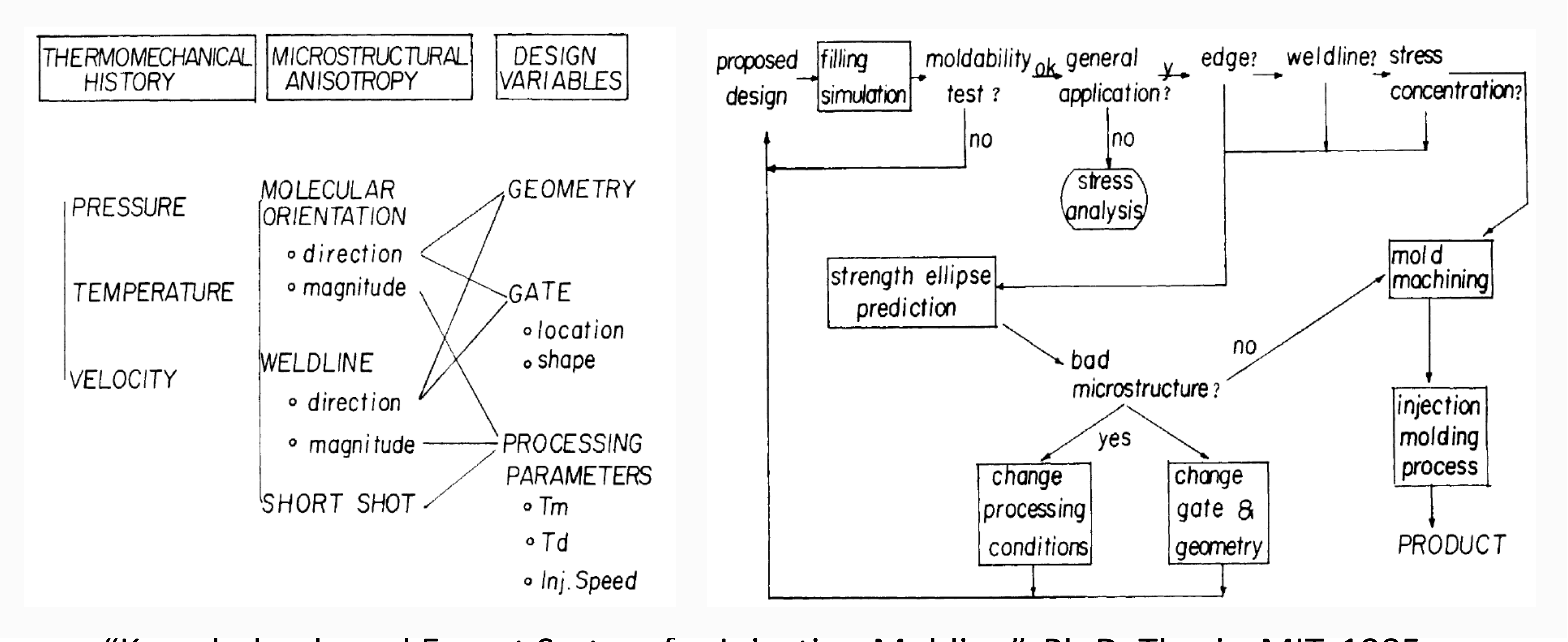
Prof. Sang-Gook Kim
MIT
Mechanical Engineering

# AI and I

- Not an AI scientist nor an AI expert.
- Need to understand the domain where AI will be applied — Manufacturing.
- Is the US dominating world manufacturing? Global Industrial Production, A national discourse.
- 1985, “Knowledge-based Expert System for Injection Molding”, Ph.D. Thesis, MIT



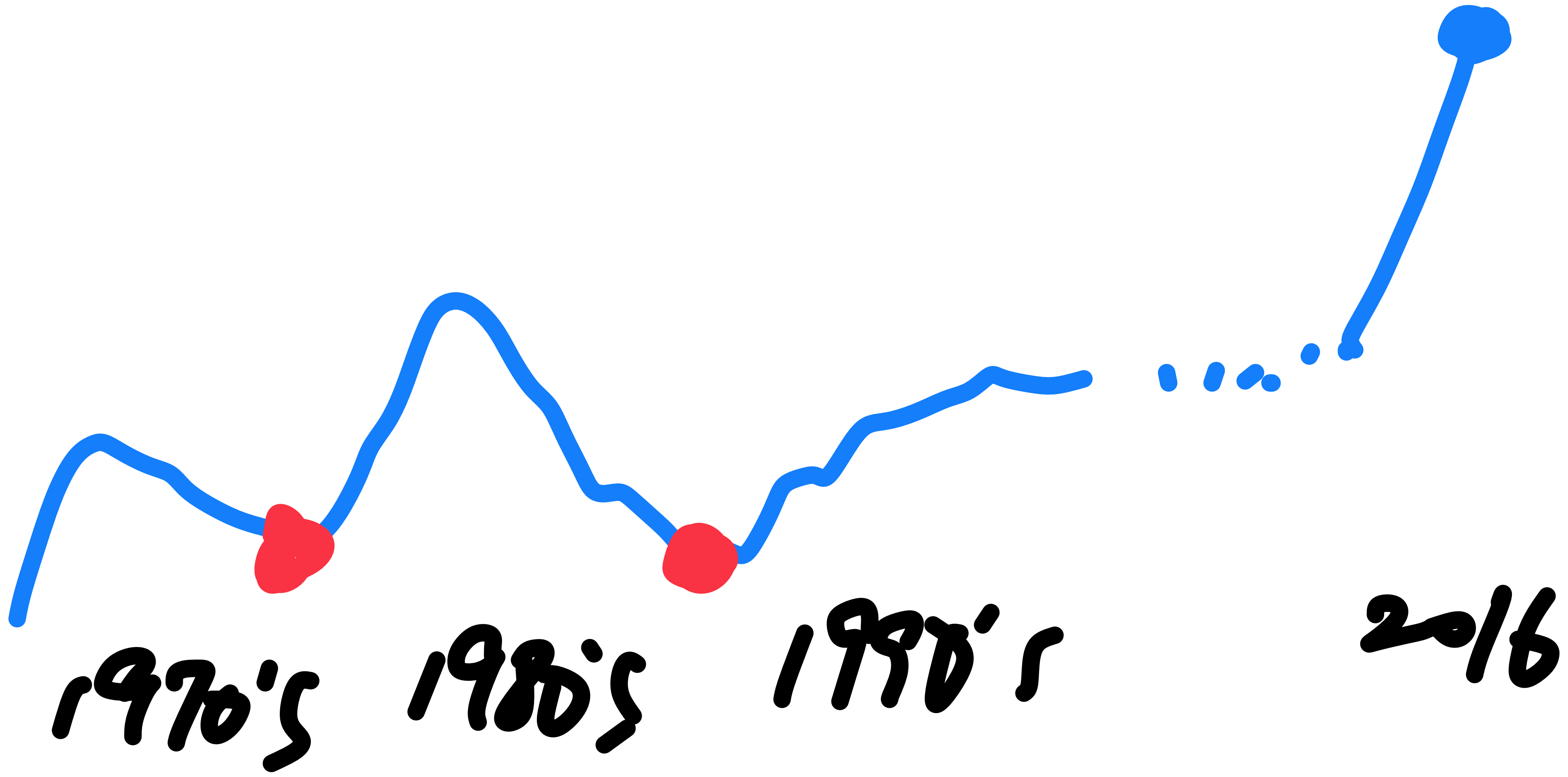
C-Flow —> Mold Flow —> Autodesk Mold Flow



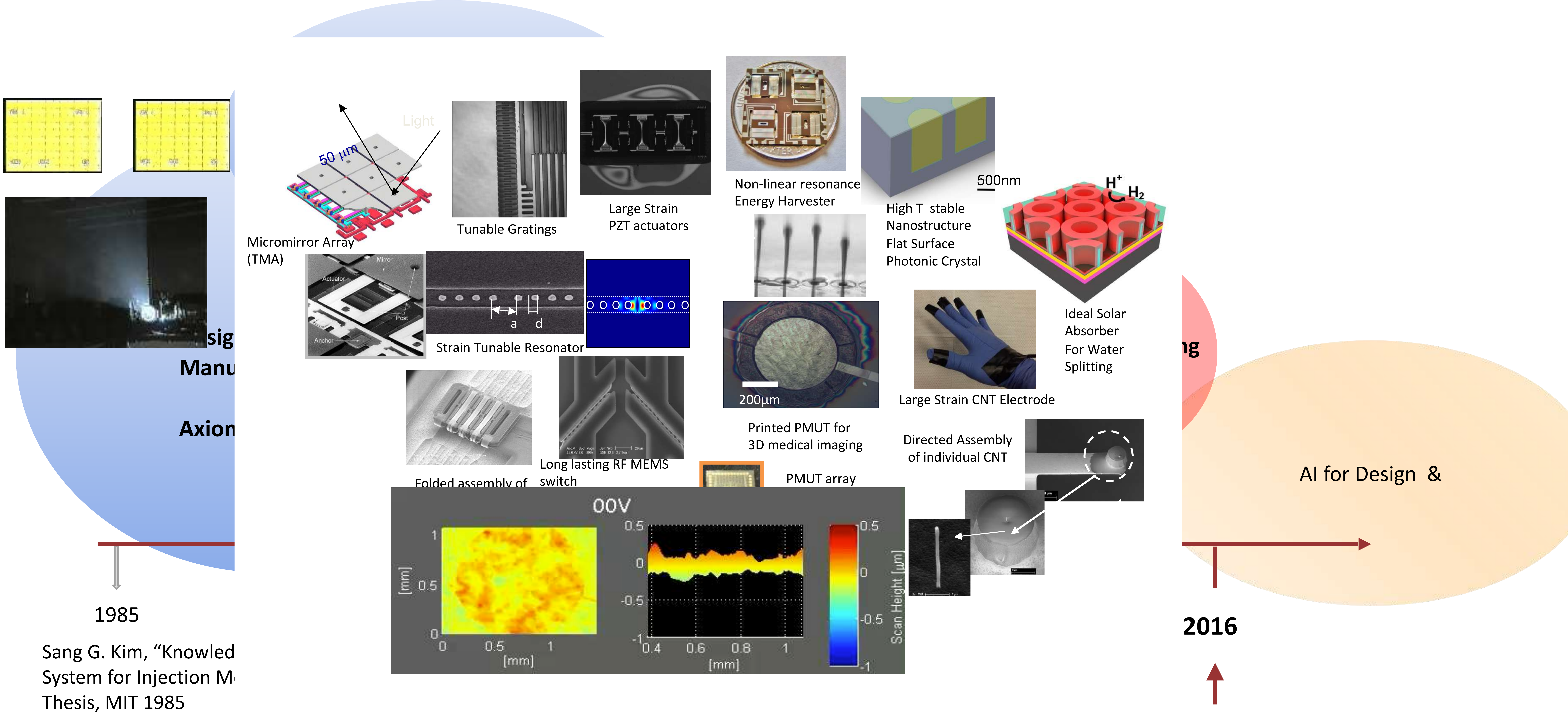

“Knowledge-based Expert System for Injection Molding”, Ph.D. Thesis, MIT, 1985

# Intelligent Manufacturing

- 1986, Int'l Conference on Intelligent Manufacturing Systems, Budapest, Hungary (Prof. Jozsef Hatvany)
- Papers on Expert Systems, Neural Network in CIRP publications, 1985-1995
  - Tool monitoring, process inspection and sensor fusion, conceptual design, CAD, process planning, process monitoring and parameter optimization, and production control among others
- AI Winters: AI research funding has had peaks and valleys from 1950s till 2010s.



# Research interests before the AI resurgence



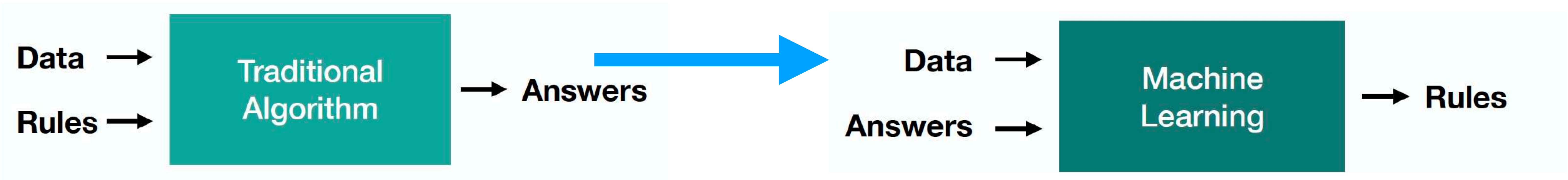

1985

2016

Sang G. Kim, “Knowled
System for Injection M
Thesis, MIT 1985

# Deep Learning and Big Data

- 2016 Oct., CBS 60 minutes, IBM's Oncological Consultant; 2016 Dec., NYT, "The Great AI Awakening*"
- 2017, "Mastering the game of Go without human knowledge" Nature 550, 354–359 (2017) Reinforcement Learning, Demis Hersabis




- 2018 Feb., MIT launches the MIT "Quest of Intelligence"
- 2019, MIT College of Computation
- 2019, Turing Award to LeCun, Hinton and Bengio
- 2024 , Novel Physics (Hinton) and Chemistry (Hassabis, Deep Mind)


Turing Award
March 27, 2019
LeCun, Hinton, Bengio

How can AI transform modern Manufacturing?

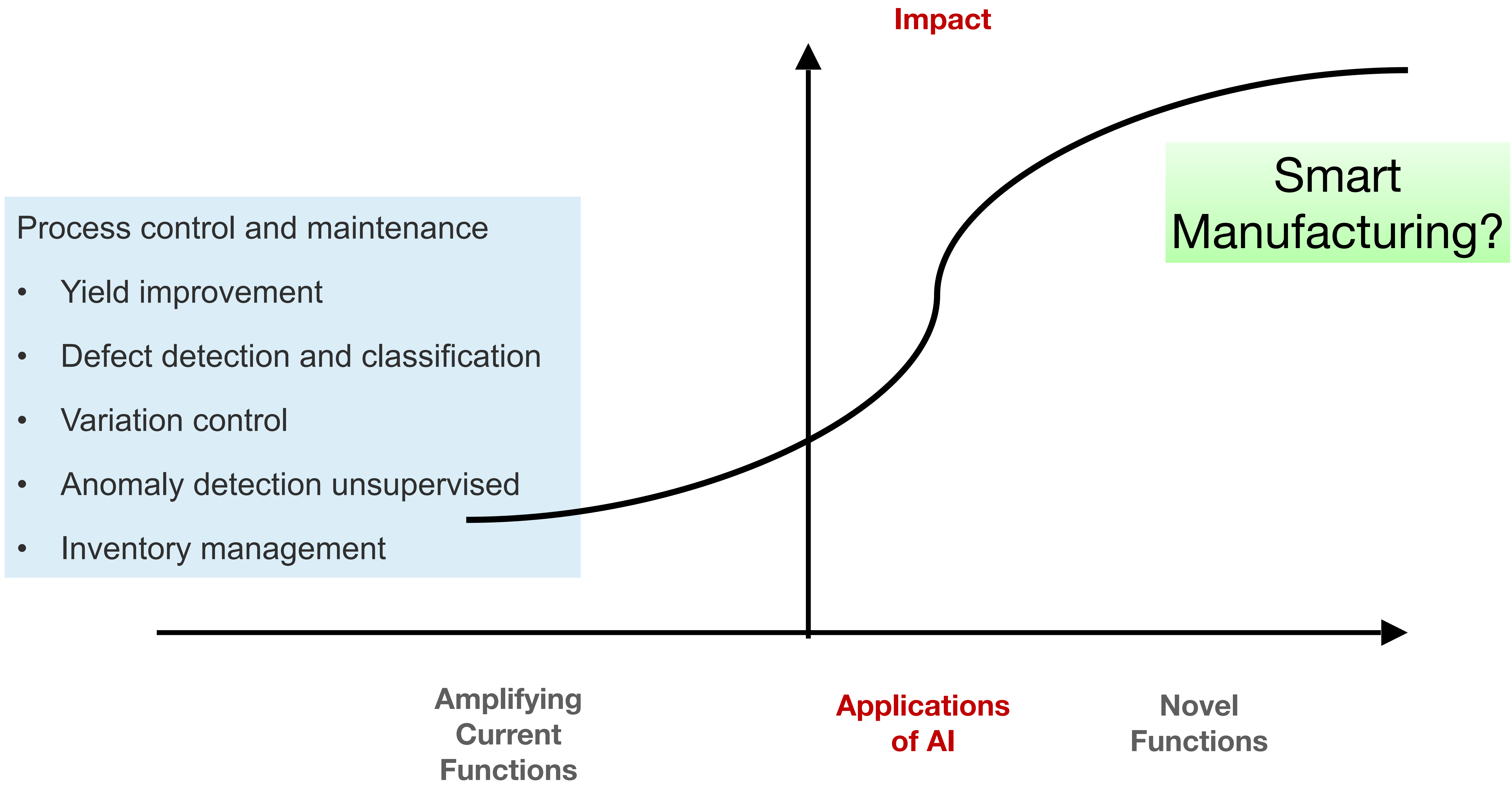
*The New York Times, Dec. 14, 2016



LeCun, Y., Bengio, Y. & Hinton, G. Deep learning. *Nature* **521**, 436–444 (2015)

# Arrival of AI in Manufacturing

- Is AI (deep learning) a game changer in manufacturing?
- True digital transformation?



# Quest for Intelligent Manufacturing (or Truly Smart Manufacturing)

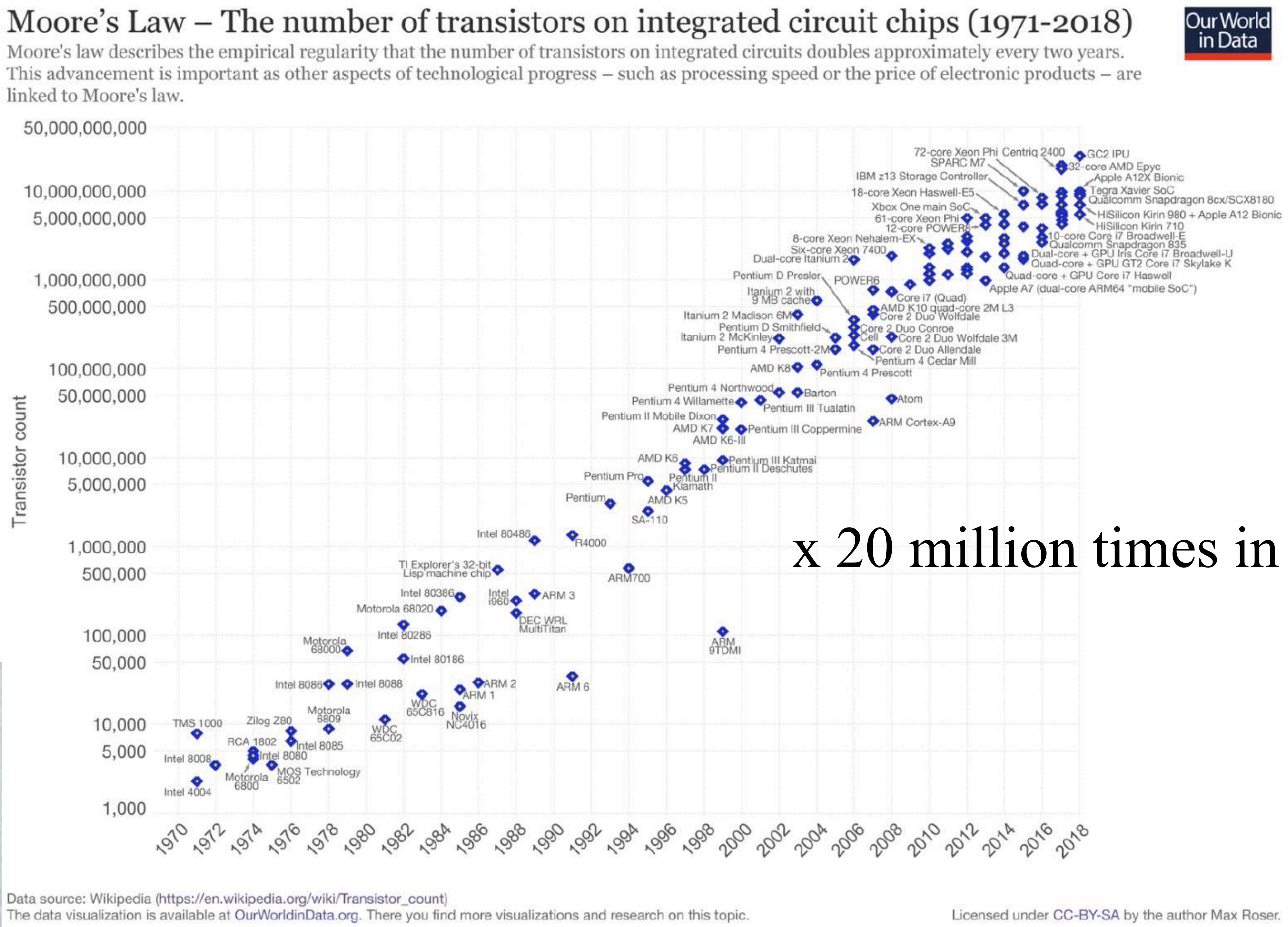
- Digitalization alone does not bring productivity edge over competitors yet.
- Entry barrier to digitalization is not high enough now.
- Industry 4.0 a hoax? No, but it is a still an **on-going thing.**
- AI will bring a real industrial revolution by integrating **human decision makers** into the Cyber Physical System loop.
- **Multimodal Modeling** help manufacturing to become smarter.

# Industrial Revolutions 4.0

- First: Steam Engine, and mechanized production, and the factory system
- Second: Electrification, Transportation, Mass Production, IC Engines
- Third: Chips, Digitalization , Communication and Automation
- Fourth: ???

- Industrial revolutions are step-change transformations in production powered by new technology.
- An industrial revolution reshapes daily life: where people live, how they work, and what “normal” family and community life look like

# Silicon Valley - Fairchild - Intel - Information Technology


x 20 million times in 50 years

https://www.dropbox.com/scl/fi/9829u0u6nh7egv3clejbj/Silicon_Valley_02_Title_1.mp4?rlkey=qfsvepx4zbo1iuskh31ljjxgn&st=45h0m7ii&dl=0

# Which Industry is at the highest AI Readiness?

- **AI Maturity Score**: data availability, digital infrastructure, talent and skills, Investment levels, regulatory environment
  - Tech, Telecommunication, Agentic AI adoption 48%
  - **Healthcare**, Financial Services,
  - **Manufacturing** ?

- **Quality Data Availability**
  - Law
  - Healthcare
  - ...
  - ...
  - Manufacturing

**AI agent** is a system that can perceive its environment, make decisions, and take actions to achieve a goal.

# Structured Data and Knowledge Availability

- Quality Data Availability
  - **Law**: rapid, widespread adoption in low value tasks, high value tasks not much (negotiation, strategy, court appearance), privacy, ethics and malpractice concerns.
  - **Healthcare**:  multimodal modeling works, privacy, ethics and malpractice concerns, open medical journals, charts privacy...
  - M**anufacturing**:
    - data hoarding, ego-driven silos, job security concerns,
    - no structured data archiving requirement (no technical papers as needed for academicians),
    - poor class imbalance problem, failure modes are rare; often only post-mortem labels exist
    - Data of one company may not be meaningful to others, but the decision rationale will be.

# Multimodal Modeling for Manufacturing

<u>Predictive Manintenance</u>: IBM, SAP, Siemens, Microsoft, GE

- **IBM-KONE**: Eelevators and escalators worldwide, IoT sensor streams plus maintenance/service logs to compute health scores and trigger proactive interventions.
- **Amazon**: Fulfillment center conveyors, vibration/temperature sensors and combined with generative AI for root-cause diagnosis and operator assistance.
- **Rolls Royce, GE**: Time-series models → detect anomalies in sensor data → extract meaning from maintenance reports to fusion layer → probability of component failure + recommended action
- **Shell, BP**: Seismic data, Pressure/flow sensors, Inspection reports (text + images), Predicts pipeline and equipment failures
-

# Multimodal Modeling for Manufacturing

High Readiness in Manufacturing:

- **Predictive maintenance**: Early fault detection from combined signals, Reduced downtime, Optimized maintenance scheduling
- **Visual quality inspection**
- **Process optimization**: Real-time parameter tuning, Drift and inefficiency detection, Improved yield and consistency
- **Worker assistance & training**: AR-based instructions, Hands-free voice interaction, Performance-based training
- **Safety monitoring**: Detect unsafe behavior, Monitor fatigue and risk factors, Proactive accident prevention

# Multimodal Modeling for Manufacturing

- **Digital Twins**: a virtual, dynamic replica of a physical asset, process, or system, stays continuously updated with real-world data.

- Supply Chain Management, Root cause analysis: **Physics informed AI**, **Physical AI**, A**utonomous Robots**

- **System of systems integration**: convergent, agile manufacturing, conflicting agents, orchestration of all agents

# Multimodal Modeling for Manufacturing

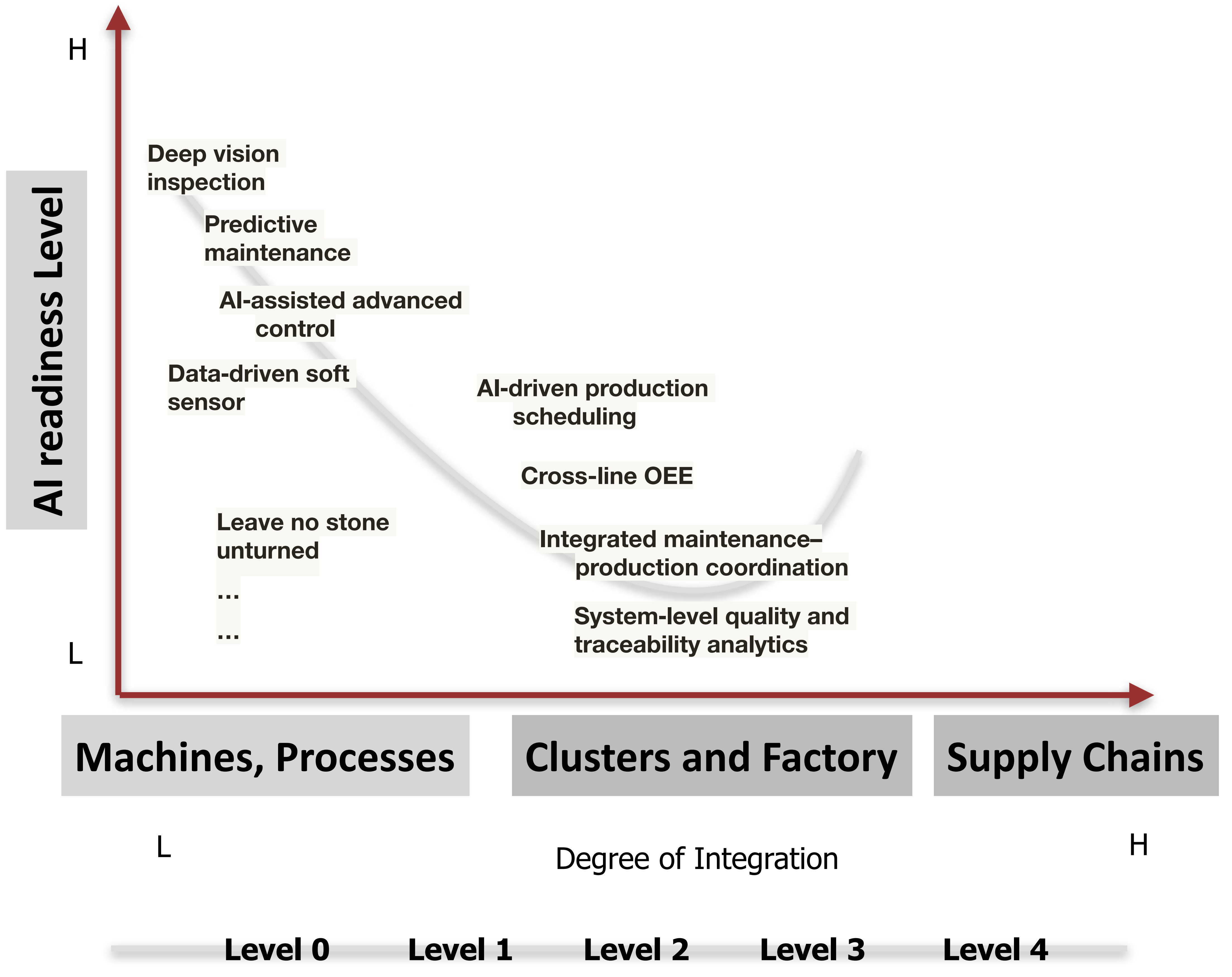
**Questions**:

- Does adding another modality meaningfully improve decisions vs. the best single-modality solution?
- Look for decisions that truly need fusion, choose a problem with current tools lack insight on the problem.
- Evaluate data alignment: Multimodal systems fail when data sources don't line up cleanly
- Is deployment realistic?
- **Cross-machine transfer or cross-plant transfer**: transferring a multimodal model from one machine to "similar" machines with minimum adaptation and re-training.
- **Foundation model** for domain adaptation, transfer learning

# Levels of Integration (enterprise business + manufacturing control)

ISA-95 (also written ANSI/ISA-95 or IEC 62264) is an international standard for **integrating enterprise (business) systems with manufacturing/control systems.**

| Level | Name | Main focus / activities |
|---|---|---|
| 0 | Physical production process | Machines, equipment, materials |
| 1 | Sensing and manipulation | Sensing and actuating |
| 2 | Monitoring and supervision | Real-time control and supervision of Level 0–1 |
| 3 | Manufacturing operations management | Scheduling, dispatching, quality, maintenance, inventory operations |
| 4 | Business planning and logistics | Long-term planning, order management, supply chain, finance |



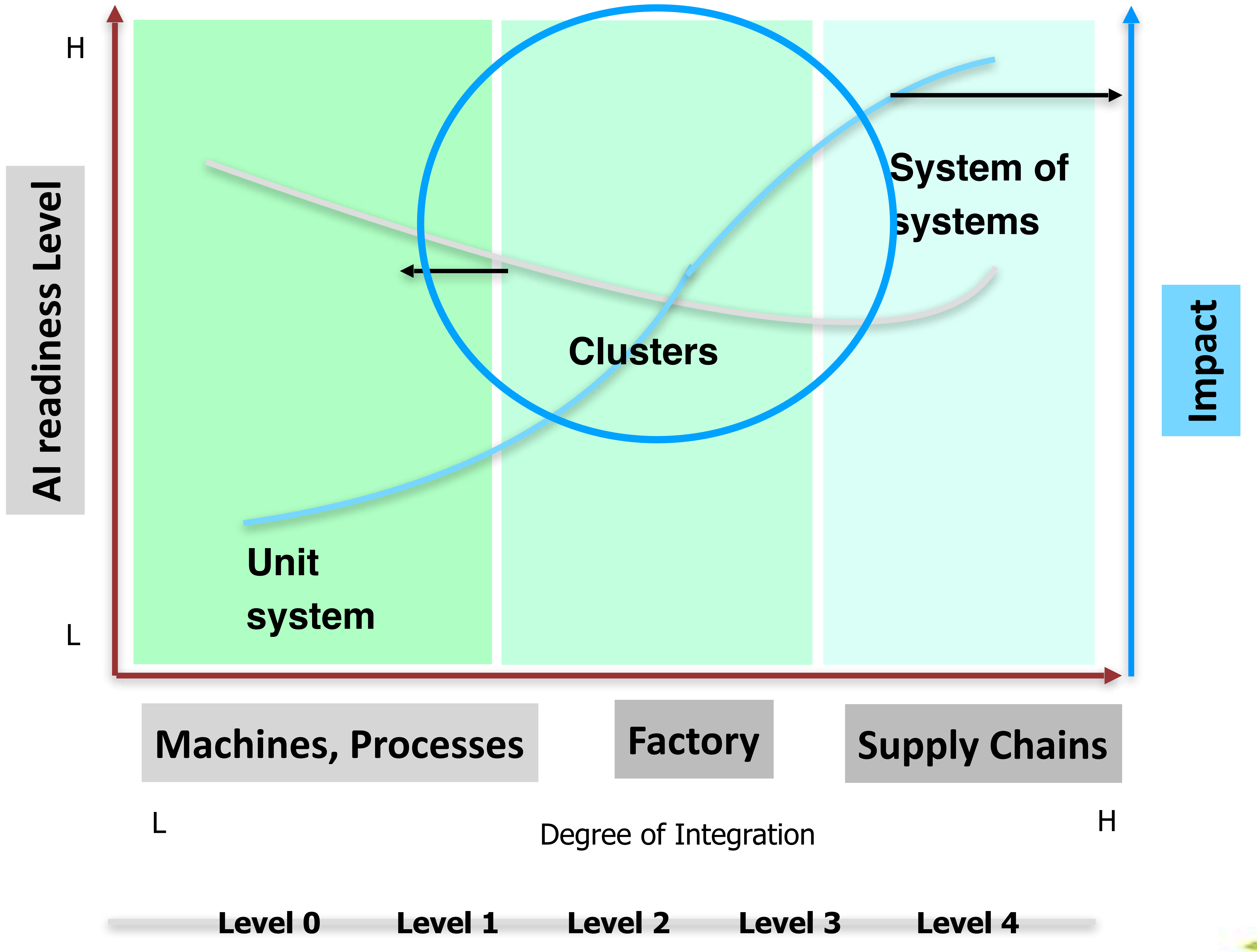
# Manufacturing Agents: System Complexity Matters

**Challenges of AI applications in  Machines and Processes Control**

- Worry about short-lived, vendor-specific AI solutions
- **Foundational framework or models**
- **Pluggable modeling** behind a stable data and process architecture

**Challenges of AI applications in Cluster Level Systems**

- Data qualty and fragmentation
- Foundational framework for **knowledge retrieval**
- Foundational framework for **systems level reasoning**
- **Data security** across divisions and supply chains
- Do not design **functionally coupled** agents.



# MIT Initiative for New Manufacturing (INM)

An MIT-wide effort uniting research, education, and partnerships to drive a future of manufacturing that is more productive, sustainable, resilient, and a source of good jobs.

*"I'm convinced that there is no more important work we can do to meet the moment and serve the nation now." -MIT President Sally Kornbluth, May 2025*

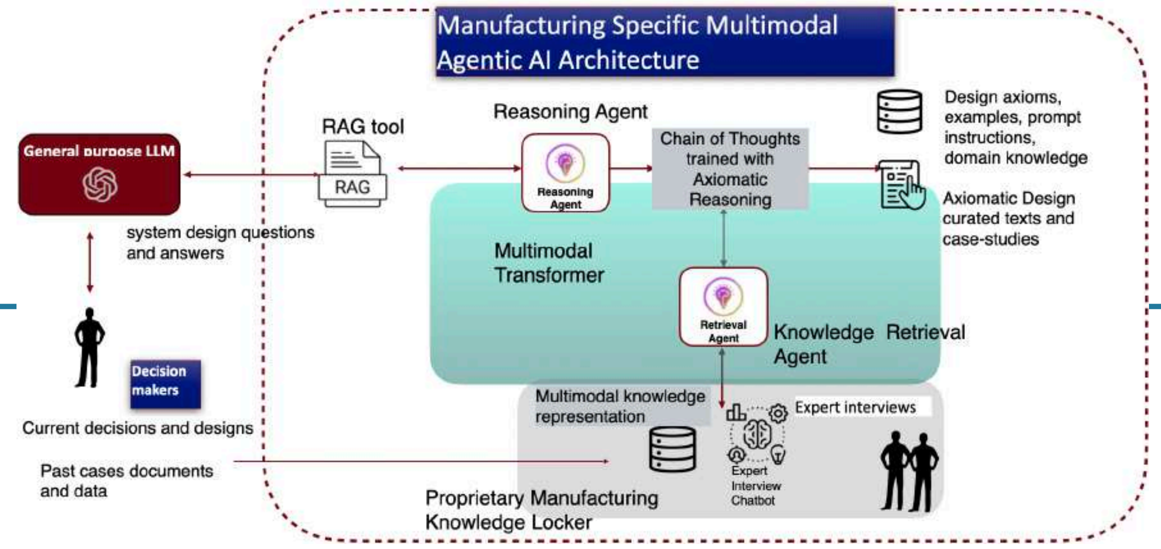
- Book "Made in America,: Regaining the Productive Edge", published in 1989, Susan Berger
- 3 Faculty co-directors: John Hart, Suzanne Berger, and Chris Love.
- MIT-Industry INM Consortium: Founding members include: Amgen, Autodesk, Flex, GE Vernova, PTC, Sanofi, Siemens
- Semiconductors ,electronics, defenseAviation, space, Biomanufacturing ,Materials, Built environment, AI for manufacturing, Machinery and automation

**Research Funding Title: Multimodal Agentic AI for Smarter Digital Twin Clusters**
**PI & Co-PI: Paul Liang, Media Lab & EECS, Sang-Gook Kim, Mechanical Eng.**
**Topic Area: Multimodal Knowledge Synthesis and Agentic AI for Manufacturing**

MIT INM
MIT Initiative for New Manufacturing

## Problem Addressed & Transformative Goal

- Digital transformation toward true Intelligent Manufacturing
- Retrieval of past decision reasonings and heuristic knowledge in multimodal form
- Contextualizing vast amounts of real-time data from sensors and IoT devices distributed throughout the system




## Team

Prof. Paul Liang (PI), a leader in the field of multimodal AI and human-AI interaction.

Prof. Sang-Gook Kim (PI), a leader in design ontology for knowledge extraction and collection.

Prof. Marvin Carl May (collaborator at NTU Singapore): a leader in complex manufacturing planning and control.

## Milestones & Risks

## Solution Concept

AIM 1: Manufacturing LLM with Multisensor Fusion and Learning

- Continuous multisensor fusion of language, vision, audio, and manufacturing-specific physical sensor information. Training a manufacturing LLM with multisensory-language instruction tuning.

AIM 2: Decision knowledge retrieval agent to build own and protected manufacturing knowledge base

- Translating *Design Thinking* ontology into a chain-of-thought (CoT) will transforming unstructured, incomplete, or poorly described heuristic and tacit knowledge into structured, curated and reusable knowledge-base.

## Outcomes Expected

1. A data loader for industrial style simulation system and sensor data, evaluation and simulation platforms for the Intel MiniFab industrial benchmark.
2. Open-source software for multisensory LLM

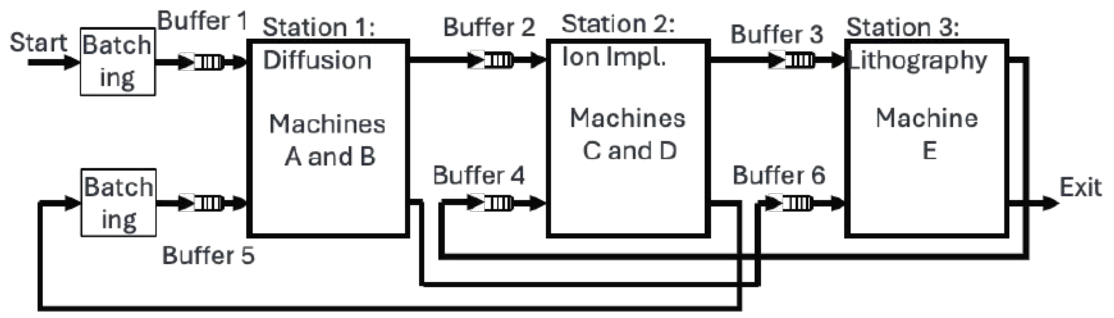
## Potential Impact

Over time, the multimodal agent will expand its capabilities through validated operations, building a long-term, self-improving knowledge base tailored to enable **truly smart manufacturing systems** across diverse industrial domains, which is the mid-term goal of our extended research.

MIT

# Foundational Platform of Manufacturing Agentic AI

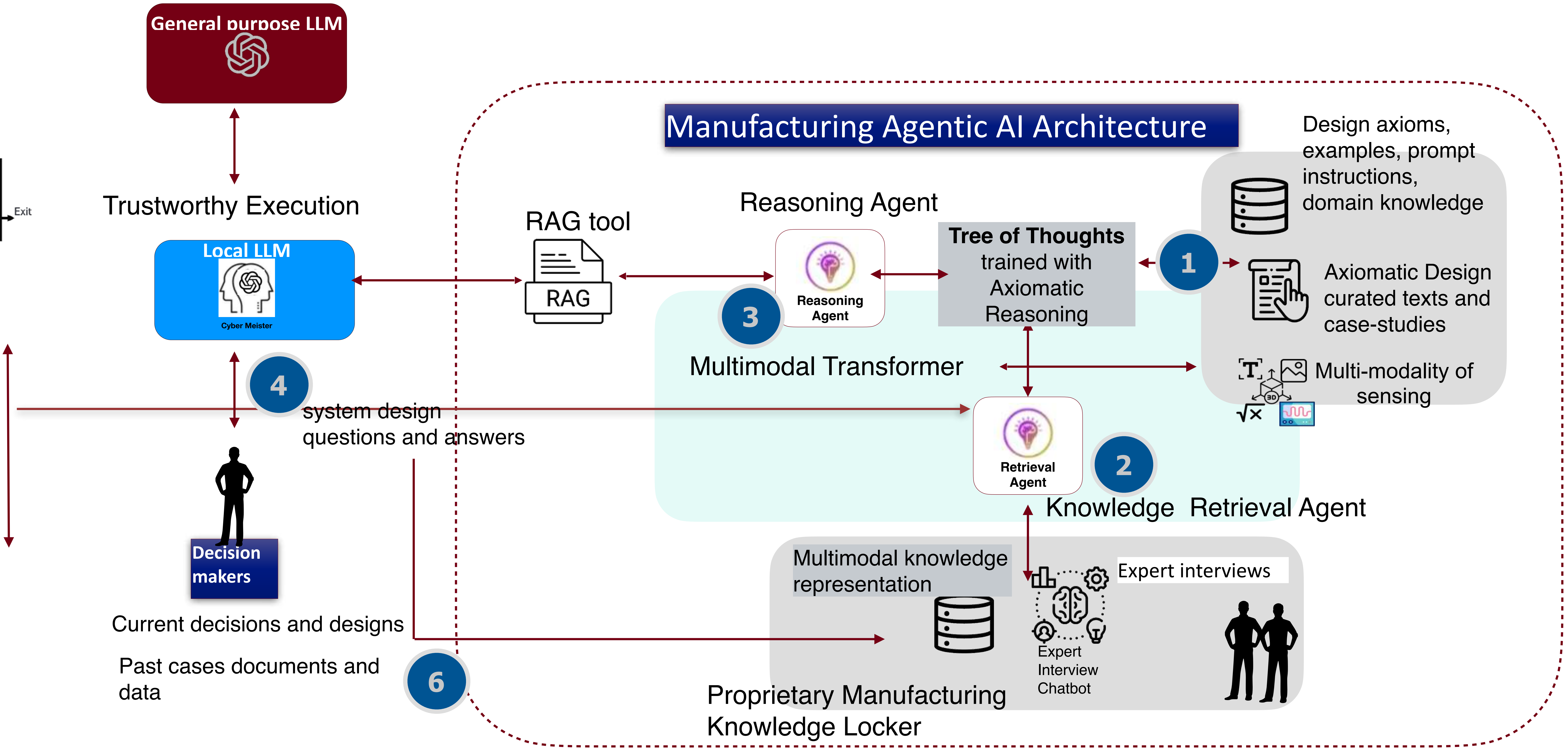
**INM project, Paul Liang and Sang-Gook Kim**

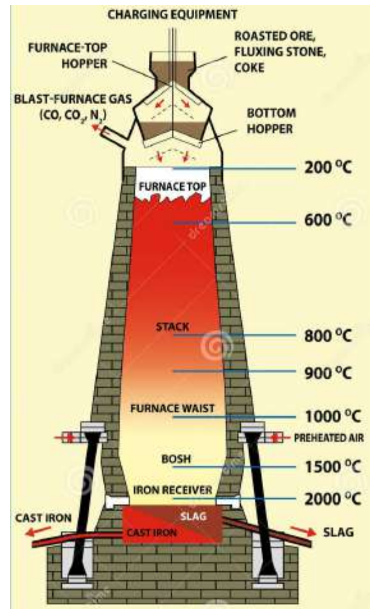



Intel MiniFab

- INM interest group systems
- Remote Machines, Factory
- 3D Metal Additive Manufactring
- Grid-level storage
- Blast Furnace

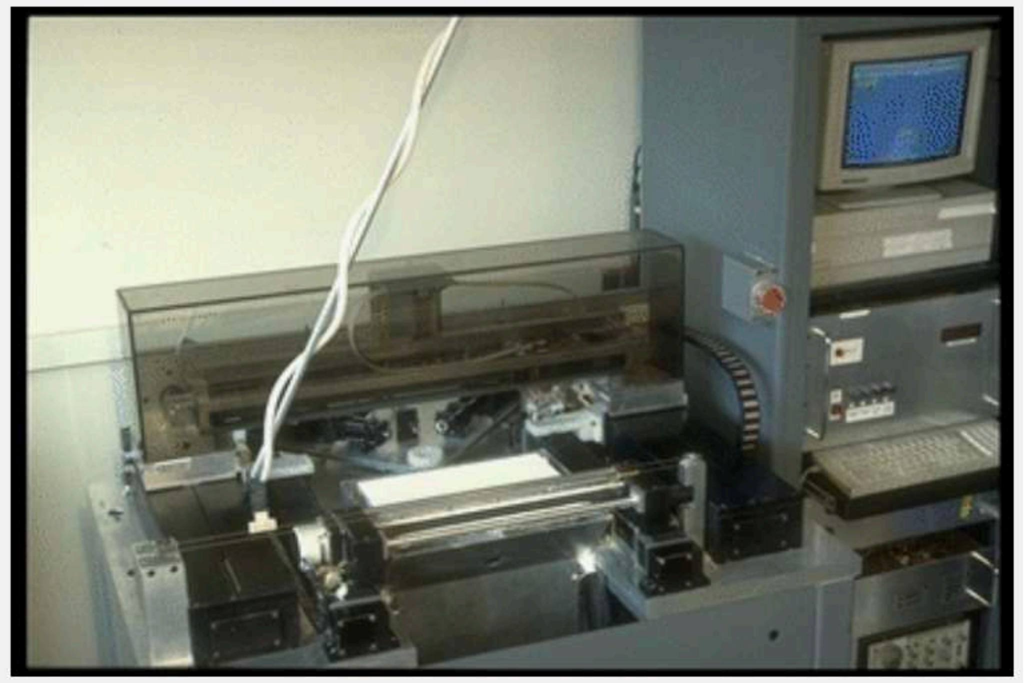
…



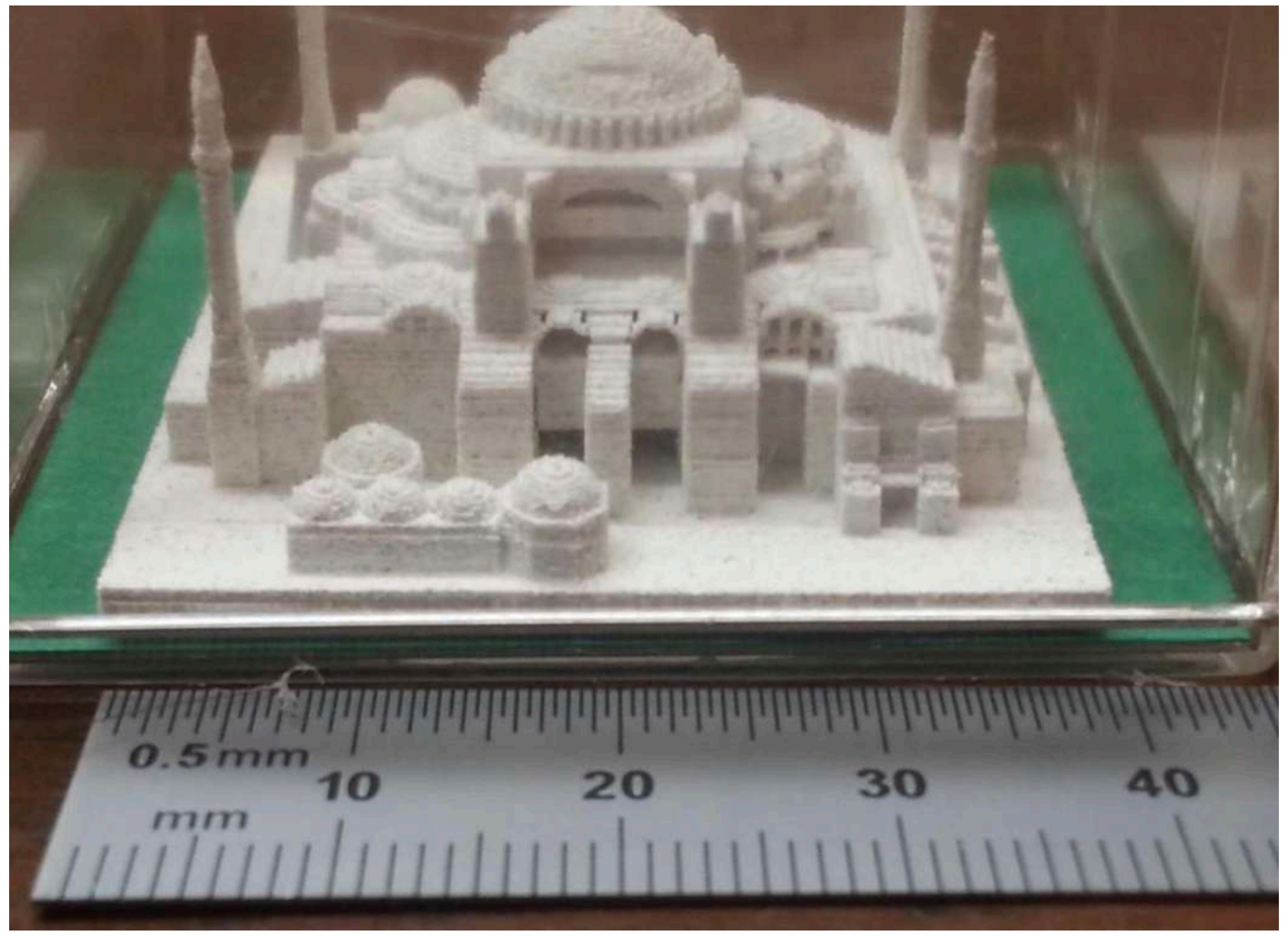
# 3D Printing - Additive Manufacturing

- In 1993, MIT professor Emanuel Sachs introduced the term “3D printing” to the lexicon, and the industry formerly known as “rapid prototyping".
- Binder jetting over dry power.

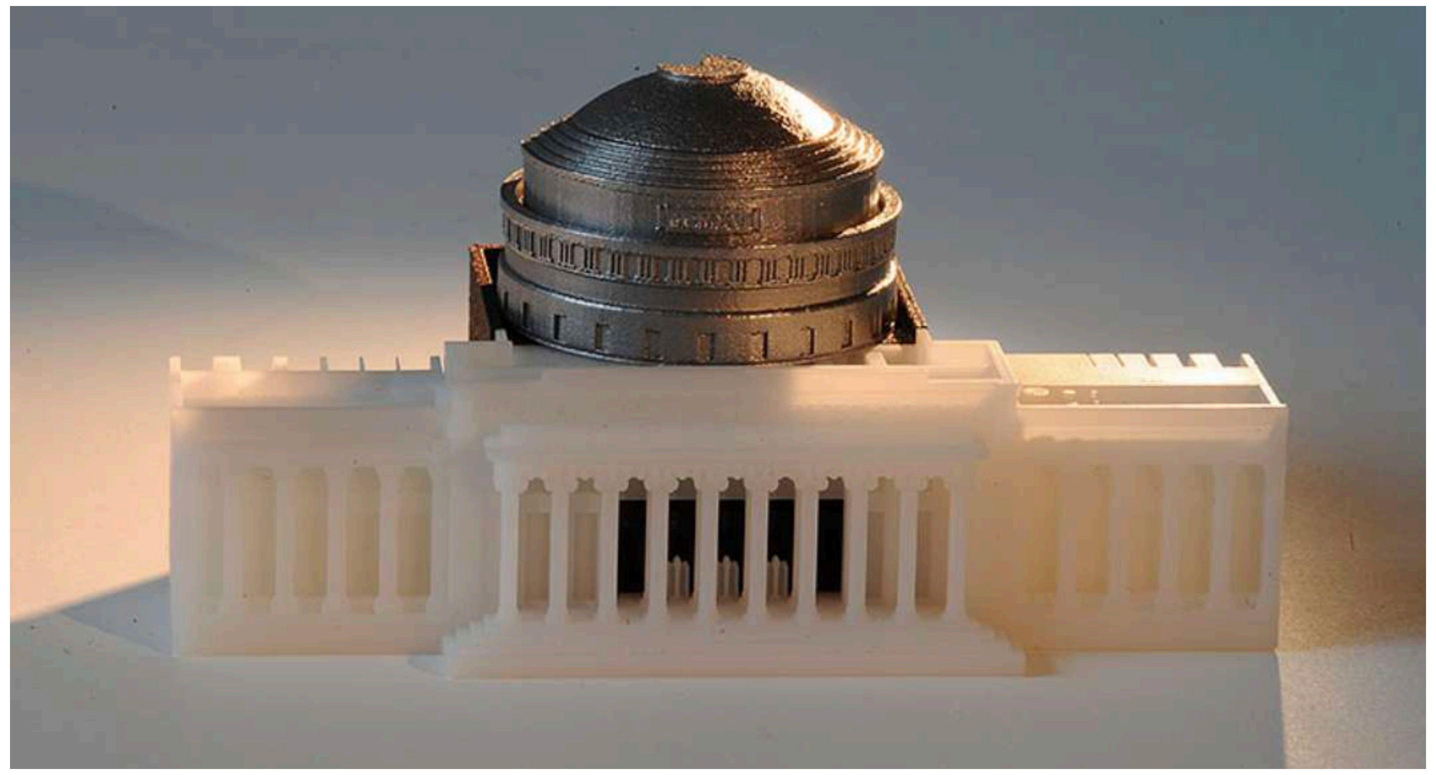


MIT Alpha 3D Printer (Image Credit: Mit.edu)

MIT Alpha 3D Printer (Image Credit: MIT Musem




3D print of the Hagia Sophia, from 1995-1996

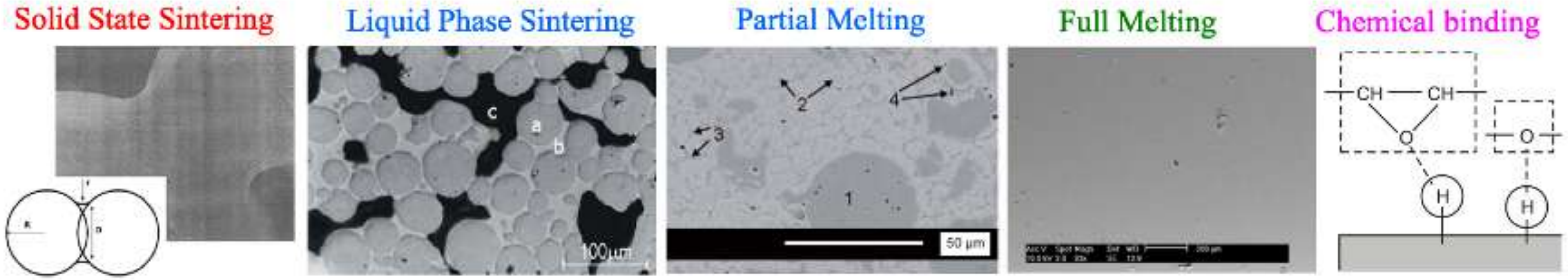


MIT’s Building 10 dome modeled after n original design by Ely Sachs and Michael Cima after their invention of binder jet printing, - F. Frankel

# Classification of binding mechanisms

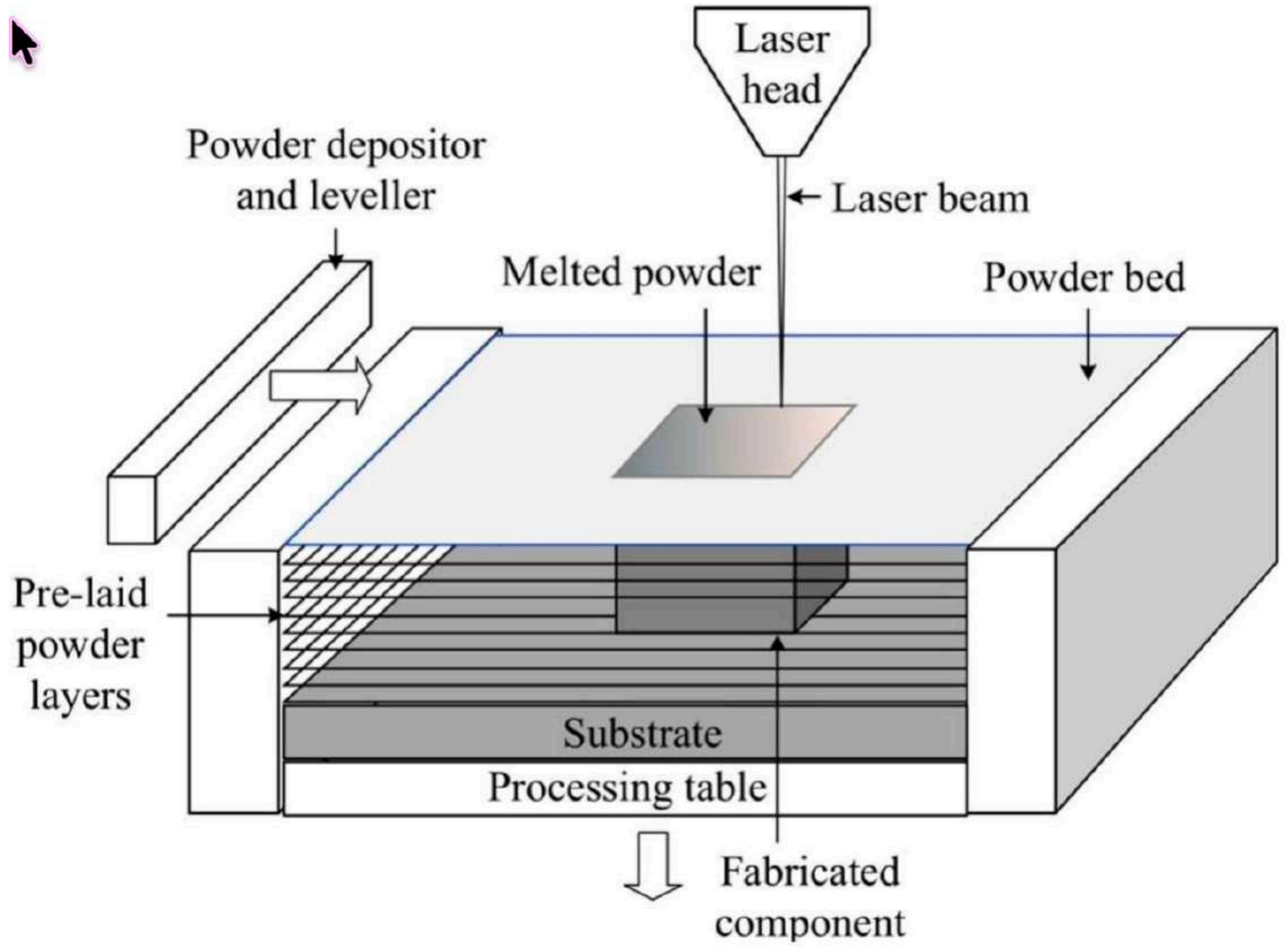
**Binding mechanism classification**

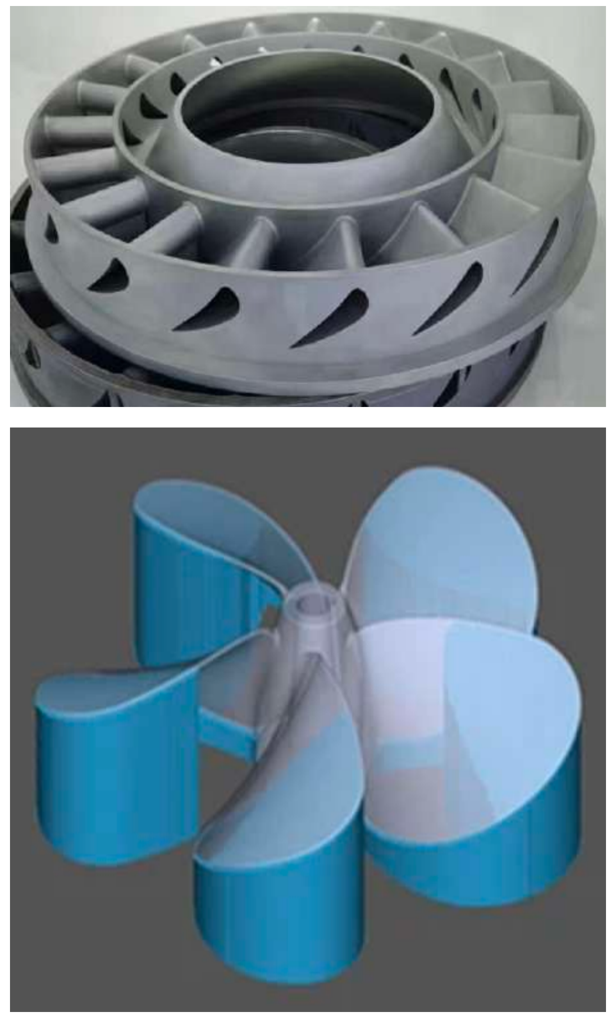
- **1. Solid State Sintering**
- **2. Liquid Phase Sintering Partial Melting**
  - 2.1 different binder and structural materials
    - 2.1.1 separate structural and binder particles
    - 2.1.2 composite particles
    - 2.1.3 coated grains particles
  - 2.2 no distinct binder and structural materials
    - 2.2.1 single phase material partially molten
    - 2.2.2 fusing powder mixture
- **3. Full Melting**
  - 3.1 single component single material
  - 3.2 single component alloyed material
  - 3.3 fusing powder mixture
- **4. Chemically Induced Binding**

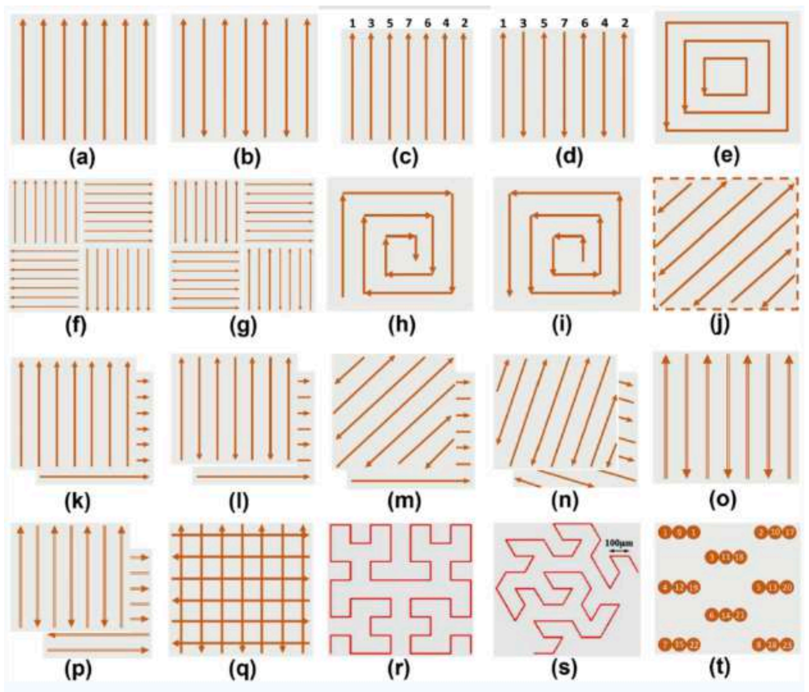


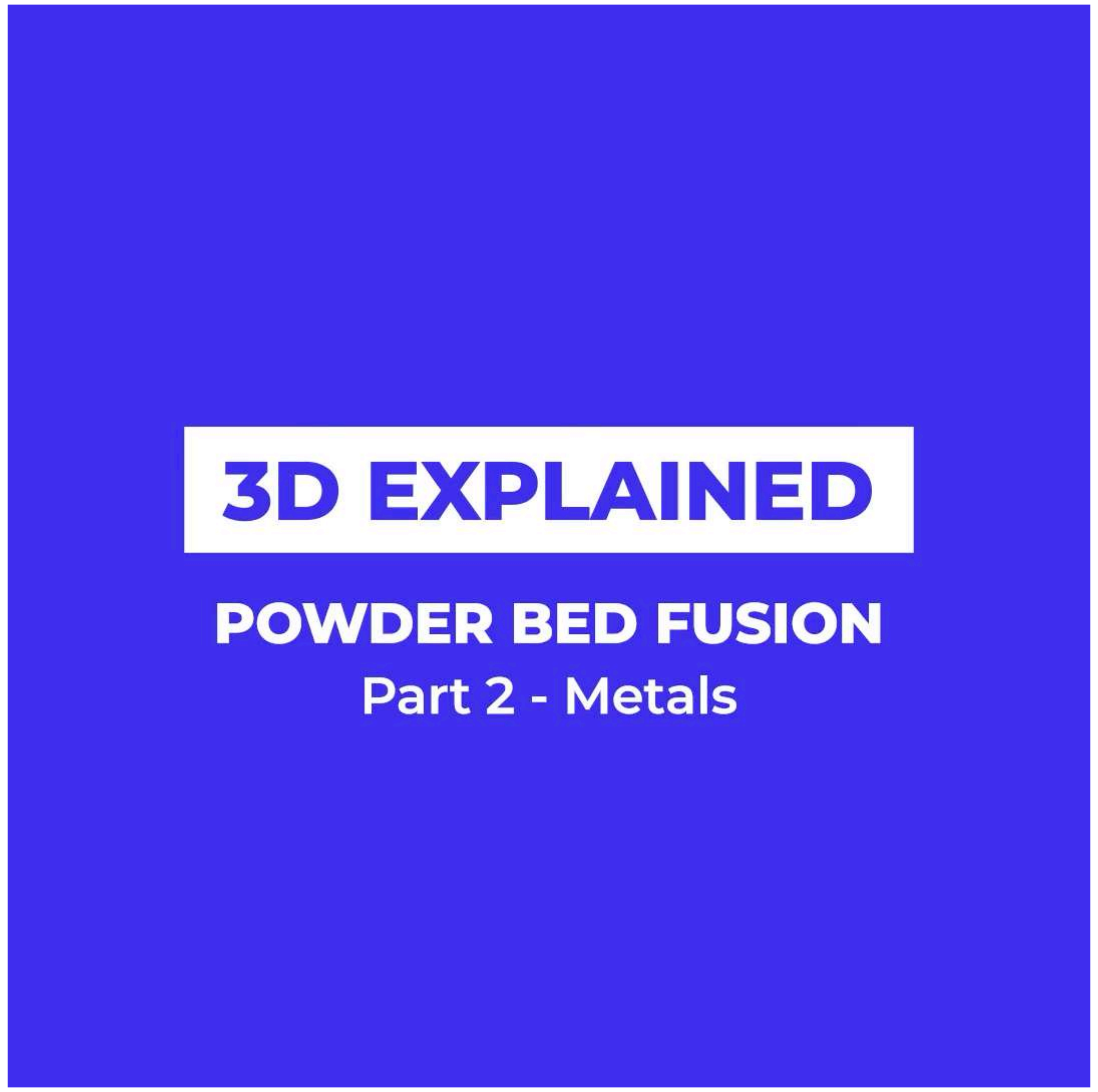
# Laser powder bed fusion (LPBF) - Metal

- Laser Powder Bed Fusion (LPBF) is a metal additive manufacturing process that uses a high-power laser to selectively melt and fuse metallic powder particles layer by layer to create complex 3D components.
- Inert Gas Chamber, Melt Pool Camera, Pyrometer, Acoustic Sensor
- Process Parameters: Laser power, Scan speed, Hatch spacing (over lap), Layer thickness
- Volumetric energy density, VED = $P / (v \times h \times t)$ [J/mm$^3$]
- Scan pattern, thermal consequences, residual stress accumulation, grain morphology, fusion quality, surface quality, dimensional accuracy








C. Zhang, Appl. Sci. 2020, 10, 81; doi:10.3390/app10010081

# Metal Additive Manufacturing

- Enables geometries you simply can’t machine
- AM metals (Al alloys, steels, Ti, Ni) with far higher stiffness for load-bearing parts if processed correctly.
- AI/ML-driven design and control are expected to defect rates and qualification time.
- Titanium alloys, Stainless steels, Nickel super alloys, Aluminum alloys

Laser powder bed fusion

The LEAP-1A engine features next generation technology for greater fuel efficiency and decreased CO2 emissions. (Image courtesy of CFM International.)



GE Aviation (GE Aerospace) fuel nozzle

Siemens high-efficient gas turbine blades

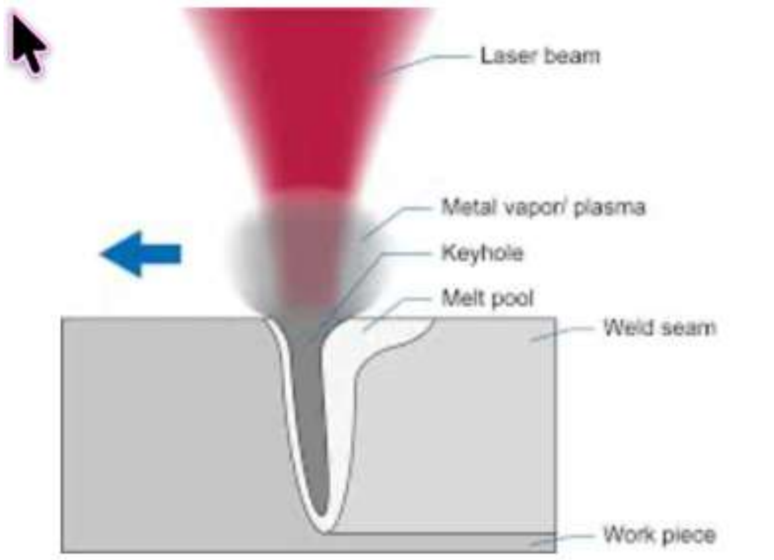
# GE Aviation, Leap Engine, Metal AM Fuel Nozzle

- A single fuel nozzle tip was reduced from about 20 pieces previously welded and brazed together to one whole piece AM.
- High fuel efficiency, 16,000 engines produced (each has 18-19 nozzles).
- Control process to eliminate defect: Keyholes, thermal distortion

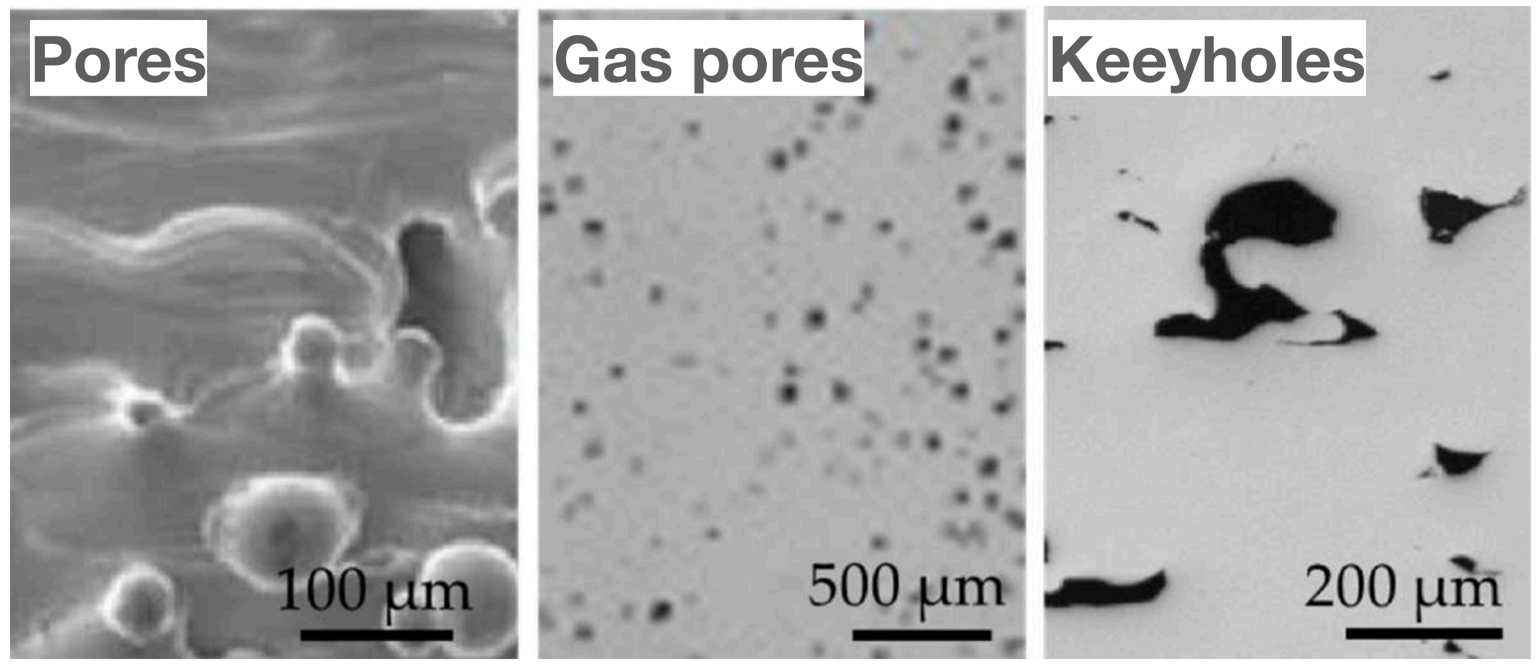
Co–Cr superalloys undergo multiple phase changes, involving transformations between FCC (γ) and HCP (ε) phases driven by rapid cooling, reheating, and residual stresses.

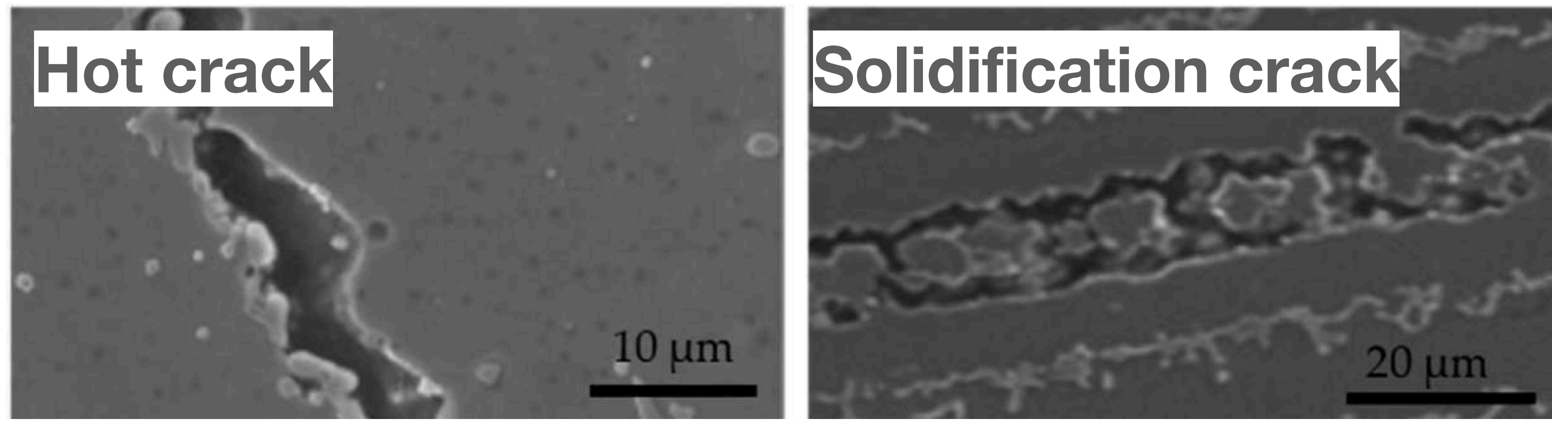
- Mel pool temperature
- Cooling rate
- Laser power, speed
- Geometry, Hatching
- Material properties

# LPBF Defects




- **Keyhole Porosity:** Excessive energy, vapor depression collapses
- **Lack of Fusion (LOF):** Insufficient energy, incomplete melting between layers
- **Gas Porosity:** Trapped gases from powder or process atmosphere
- **Surface Roughness:** Partially melted particles, balling effect
- **Spatter:** Ejected molten droplets redeposit creating defects
- **Microcracks:** Thermal stresses exceed material strength
- **Delamination:** Poor interlayer bonding, residual stress, distortion



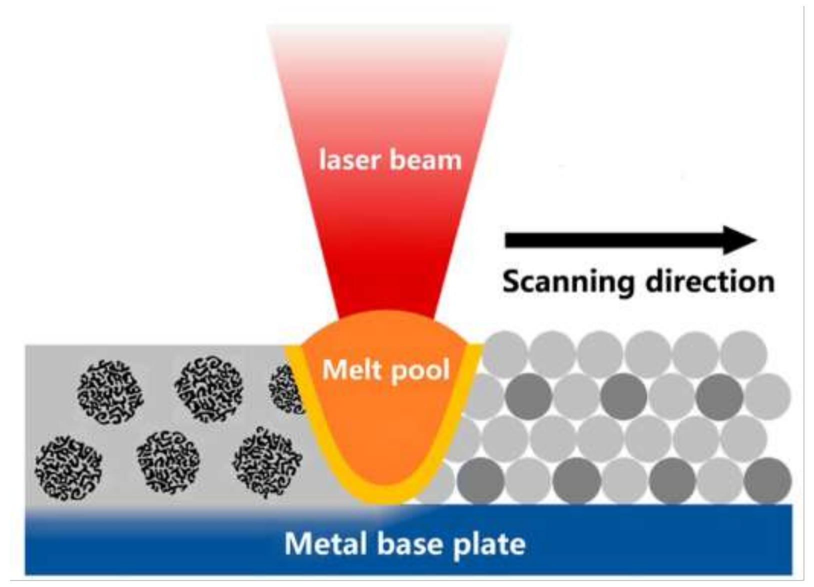




**How to achieve Process Repeatability, Quality, Build Rate**

**AL-enabled Multimodal Modeling Agents?**

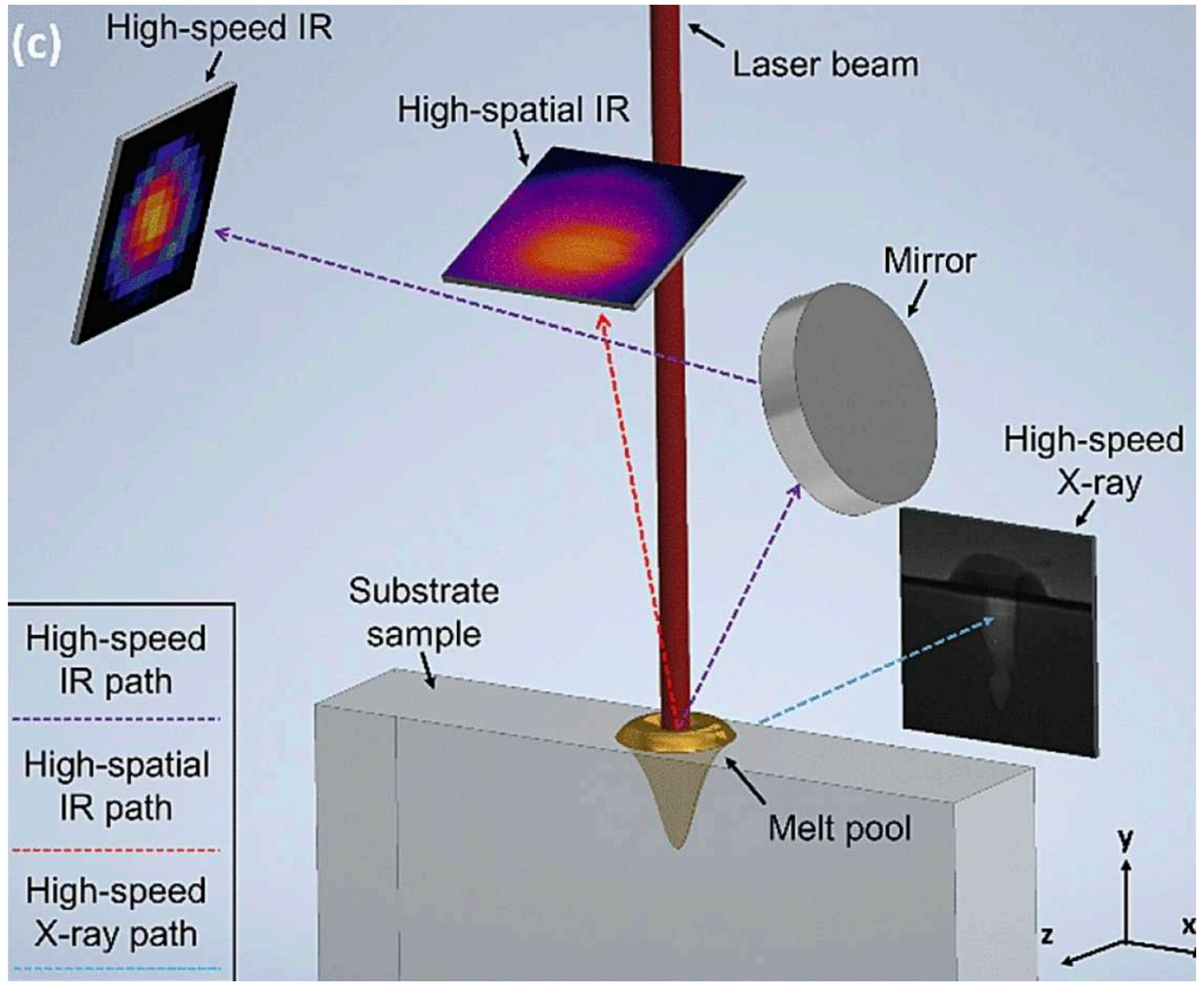
# Melt Pool Physics

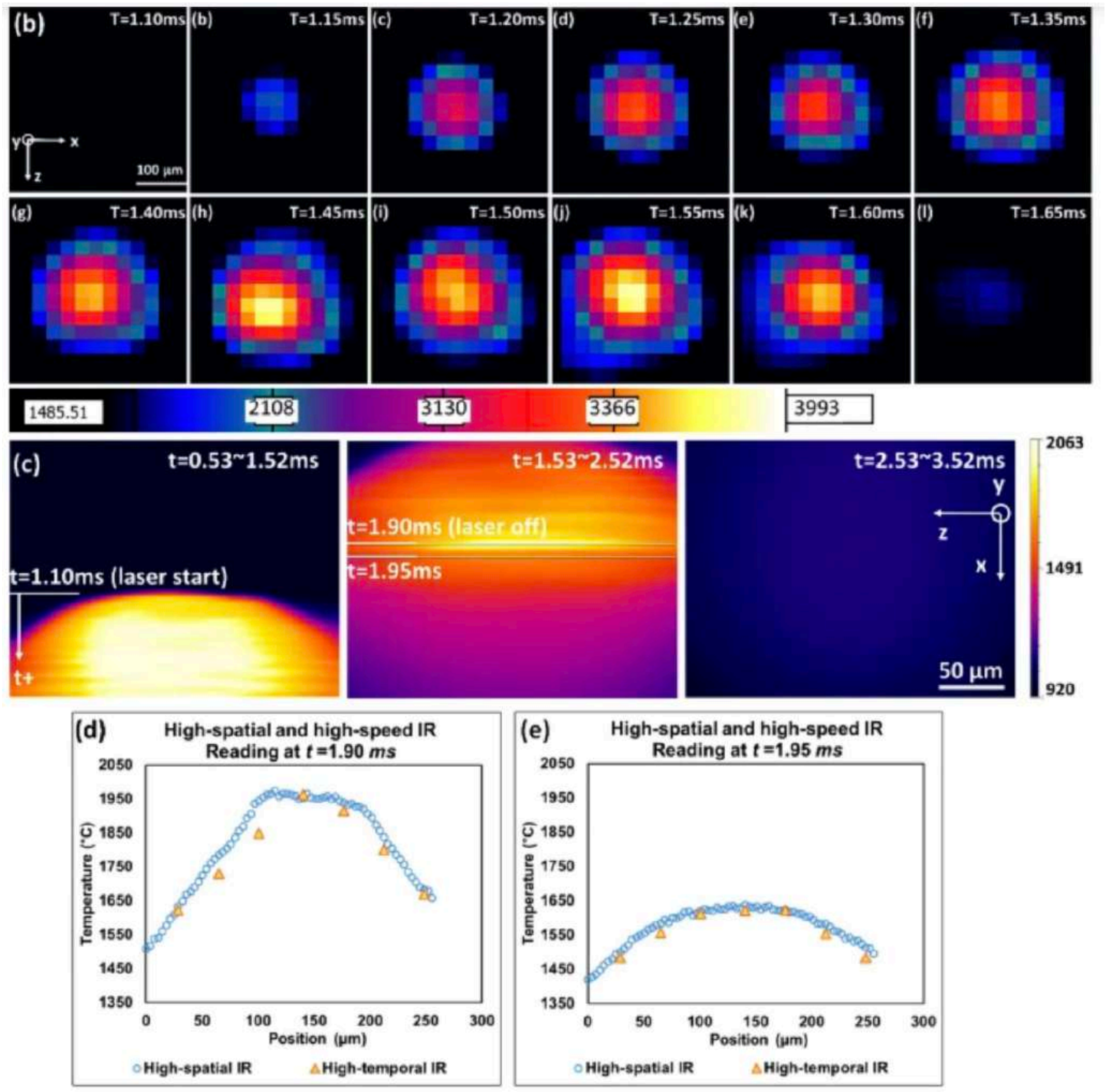
- **Extreme Thermal Gradients:** $10^5$-$10^6$ K/m creates rapid solidification
- **Marangoni Flow:** Surface tension gradients drive complex fluid circulation
- **Recoil Pressure:** Vapor pressure from evaporation affects pool shape
- **Melt Pool Dimensions:** 50-200 μm width, 50-300 μm depth




- Synchrotron X-ray imaging: directly image melt pool shape, depth, keyhole cavity, and pore dynamics in spot melts
- High-speed IR camera (tens of kHz, ~30 μm/pixel)for overall melt pool/plume temperature evolution.
- High-spatial-resolution IR camera (~3.6 μm/pixel, ~1 kHz) capturing detailed surface temperature fields

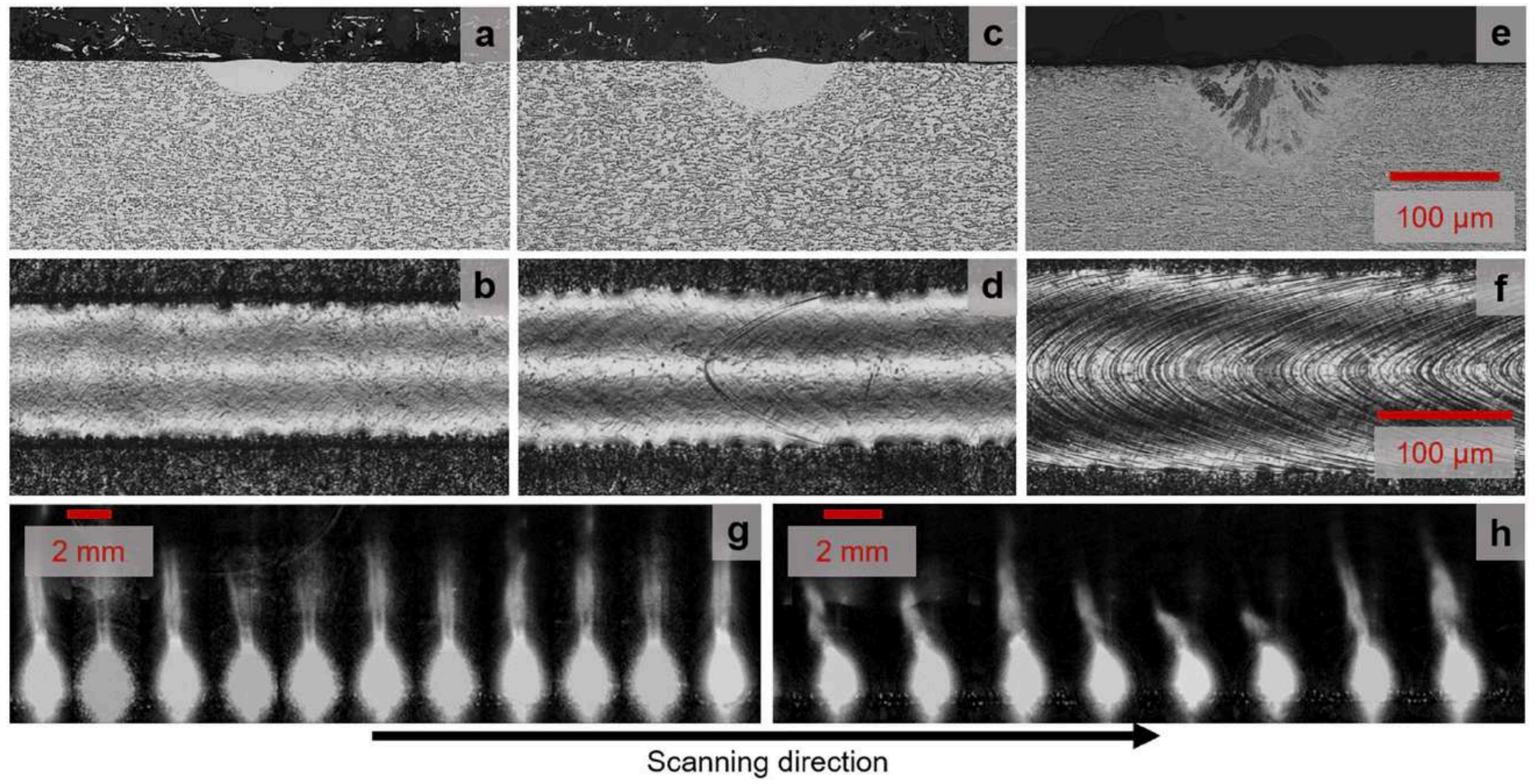
Argonne N Lab

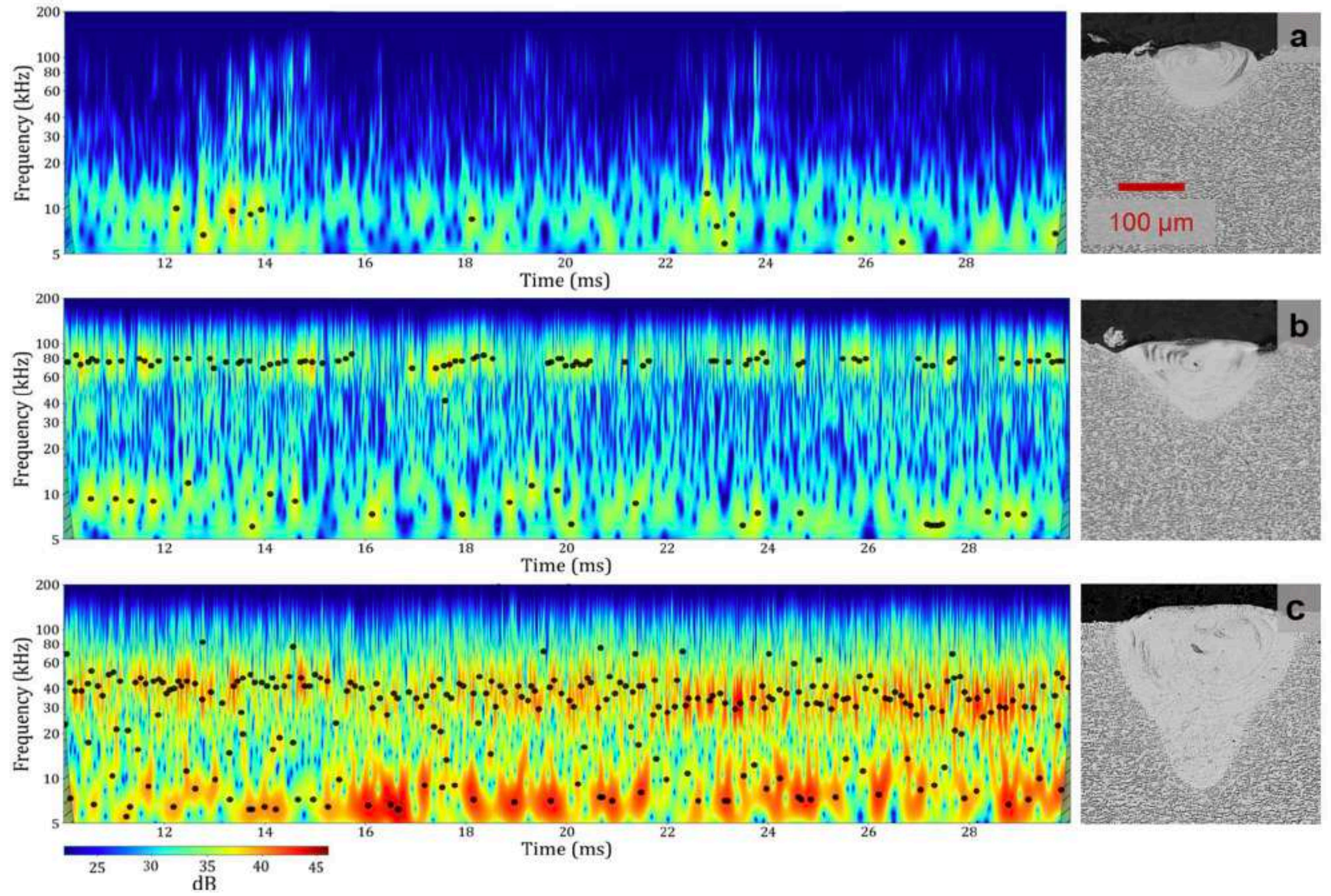


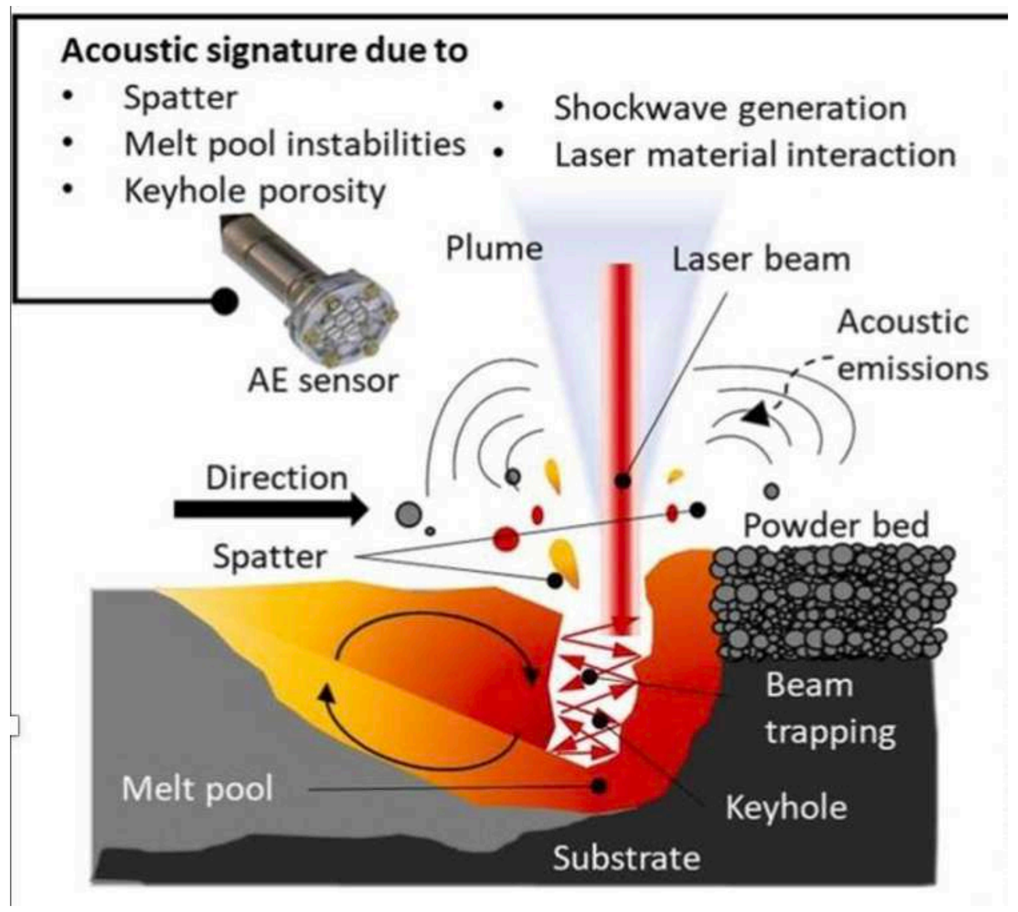




In situ melt pool measurements, Argonne N Lab, Wang, R.. *et al.* In situ melt pool measurements for laser powder bed fusion using multi sensing and correlation analysis. *Sci Rep* 12, 13716 (2022).

# Acoustic emission monitoring reveals keyhole formation



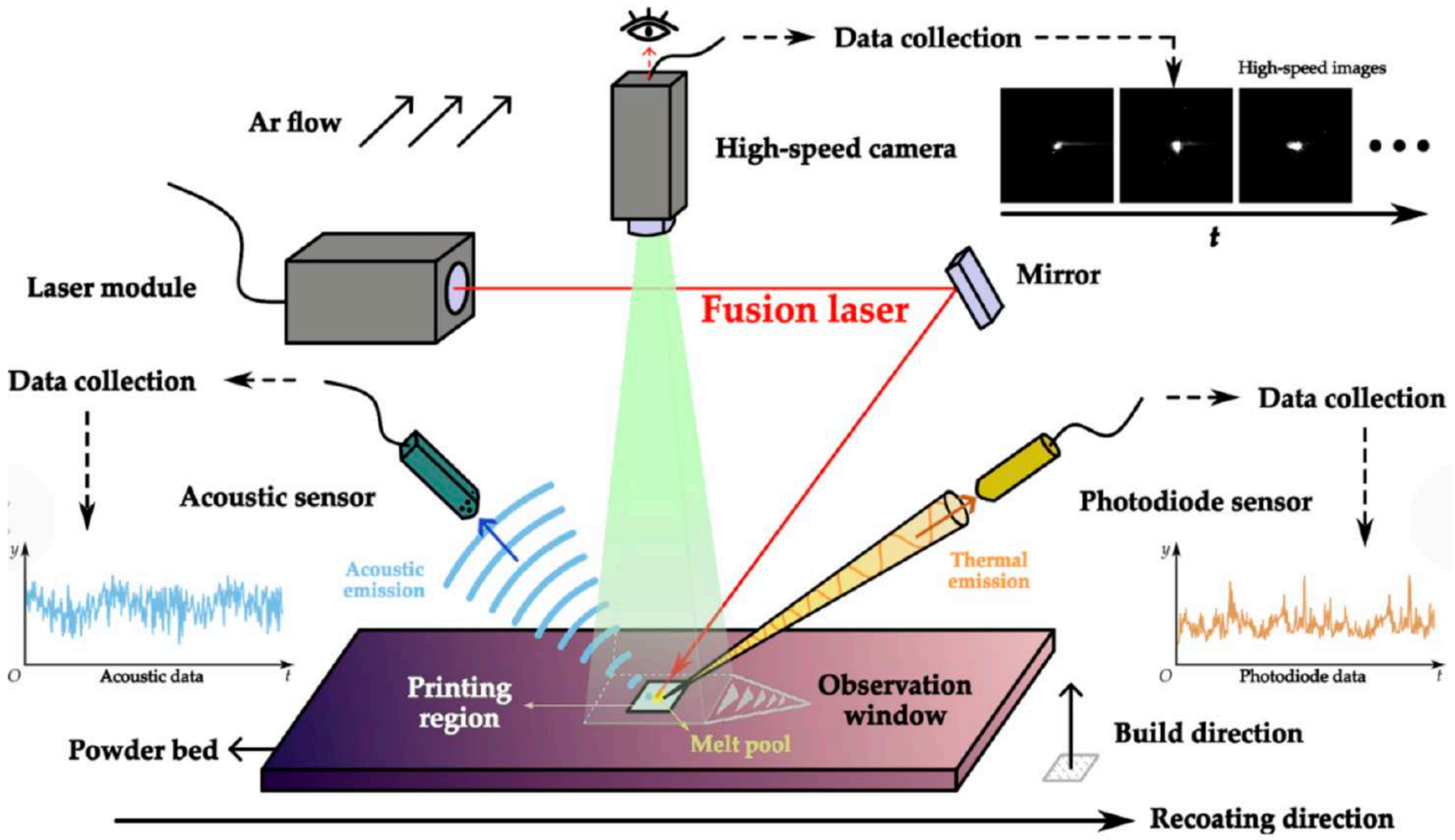




L. Galiègue et al., Nature Scientific Report , 2025, I https://doi.org/10.1038/s41598-025-27232-1

- Link sound signatures with conduction vs. keyhole melting and porosity
- Its frequency is inversely related to melt pool depth
- The oscillations of the liquid–vapor interface and plume can serve as an indicator of keyhole evolution and porosity risk.
- The liquid/vapor interface (keyhole wall + plume) oscillates because it is a dynamic balance of competing forces: laser energy and vapor pressure <—> surface tension and liquid inertia

# MM fusion steps and future directions




- Multiple sensors (optical cameras, photodiodes, IR, acoustic emission, process logs)
- Aligns data in space and time so that each voxel/scan vector has synchronized signals.
- Uses representation-level fusion with separate encoders.
- Uses feature-level fusion by extracting features per modality (textures, intensity, spectra, temporal descriptors), concatenating into a joint feature vector.
- Uses decision-level fusion where individual modality-specific models output defect/quality scores.
- Improves state/defect classification under changing process parameters and geometries.
- Enables domain adaptation and transfer learning so models trained on one machine/parameter map can be adapted to others using fused features and unsupervised or semi-supervised techniques.

# On going project, INM

# Trumpf Se Co, KG

Germany, founded 1923, strong, innovative medium-to-large family businesses, ~18,000 employees, 80+ countries

**Machine tools**
Equipment for shaping and cutting metal
Includes:
- Laser cutting machines
- Punching machines
- Bending systems

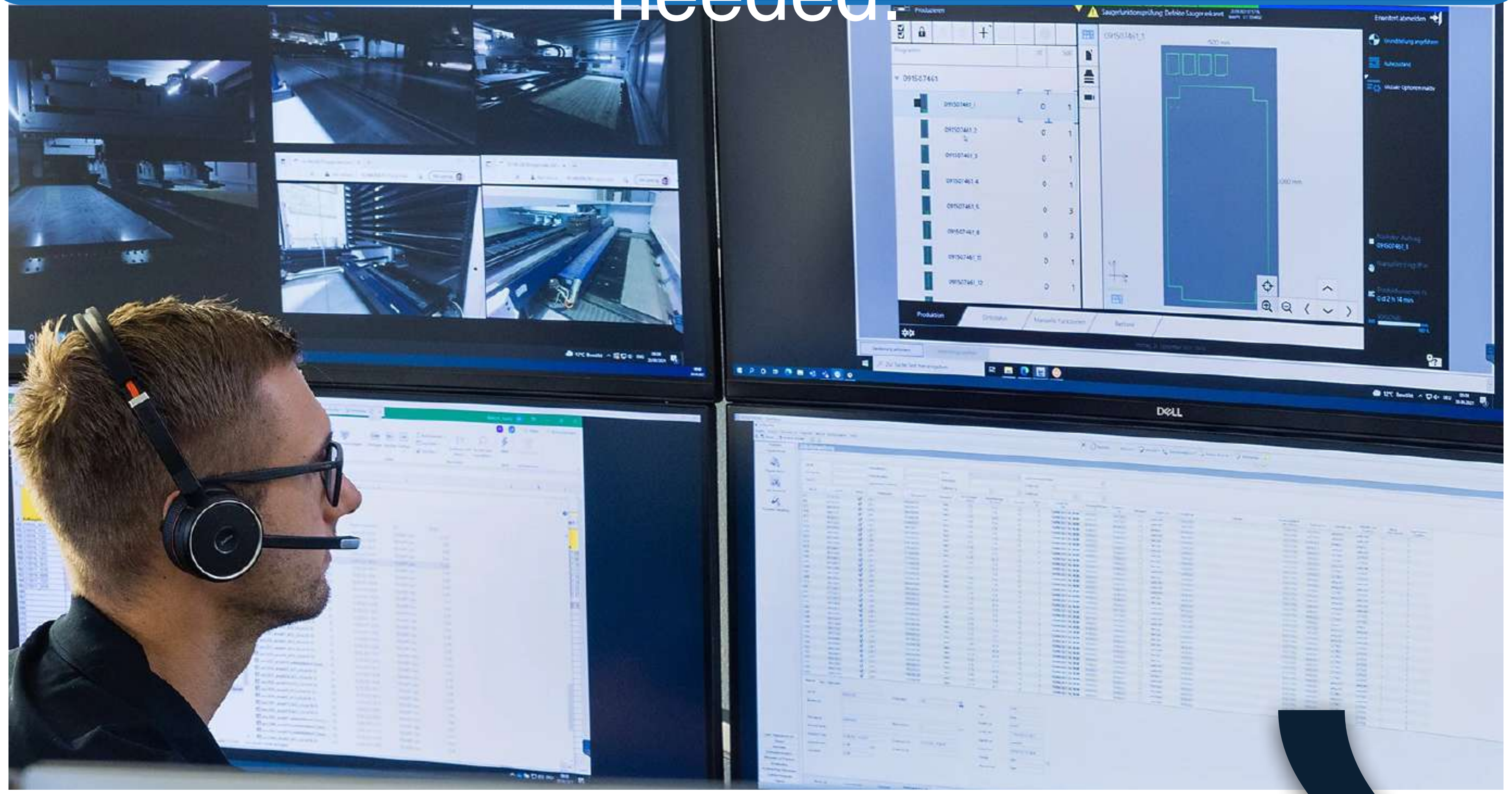
**Laser technology**
•Industrial lasers for:
- Cutting and welding materials
- Semiconductor manufacturing (chips)
- Medical devices



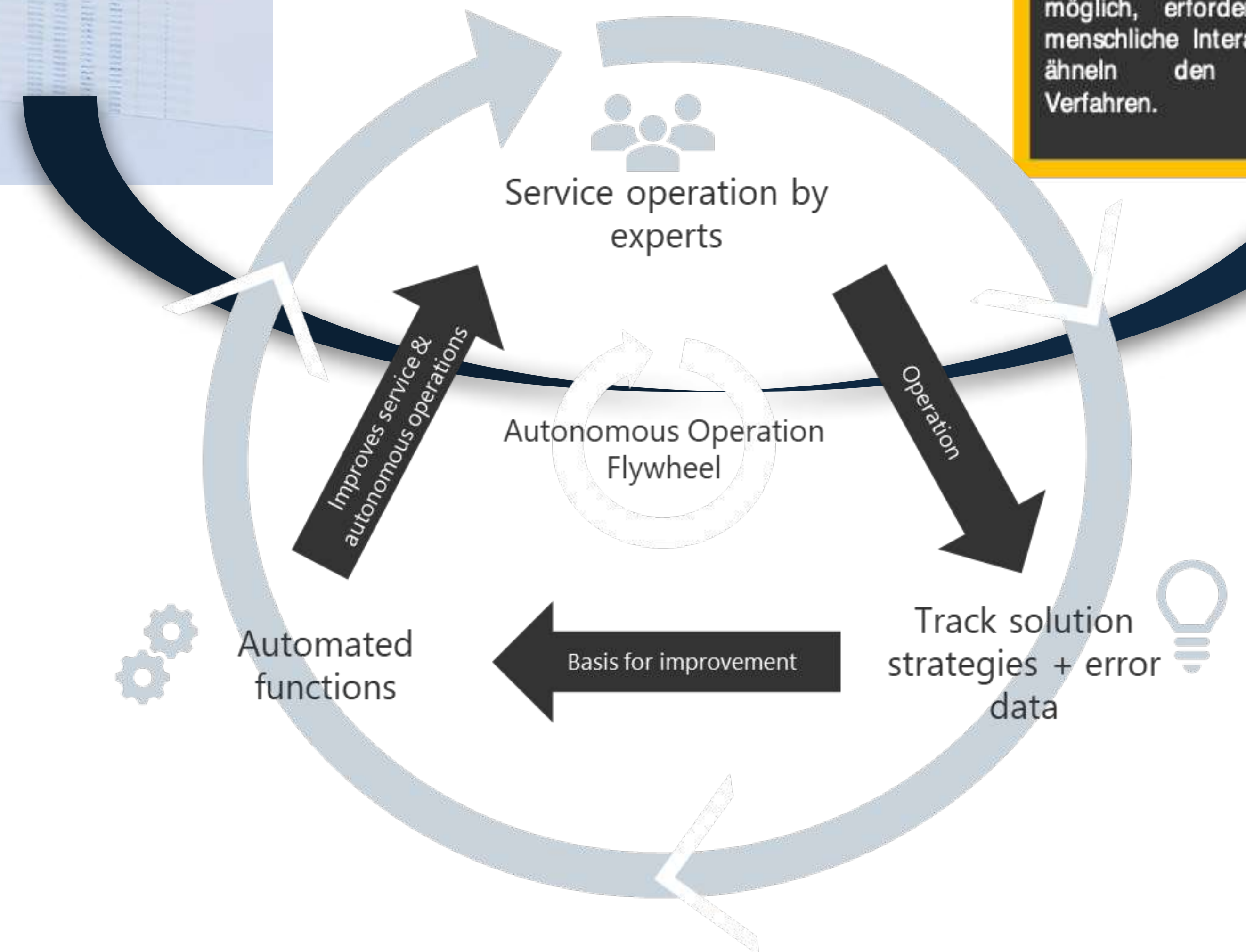
MITMECHE

# Remote operator as starting point -> Goal: automatic troubleshooting

**Today:** Remote Operator extensive skills and comprehensive training is needed.



**Vision:** Machine can eliminate faults itself.


Confidential

# Machining breakdown

## Multi-modal data from real machining operations and breakdowns

### System visualization (Real-Data)

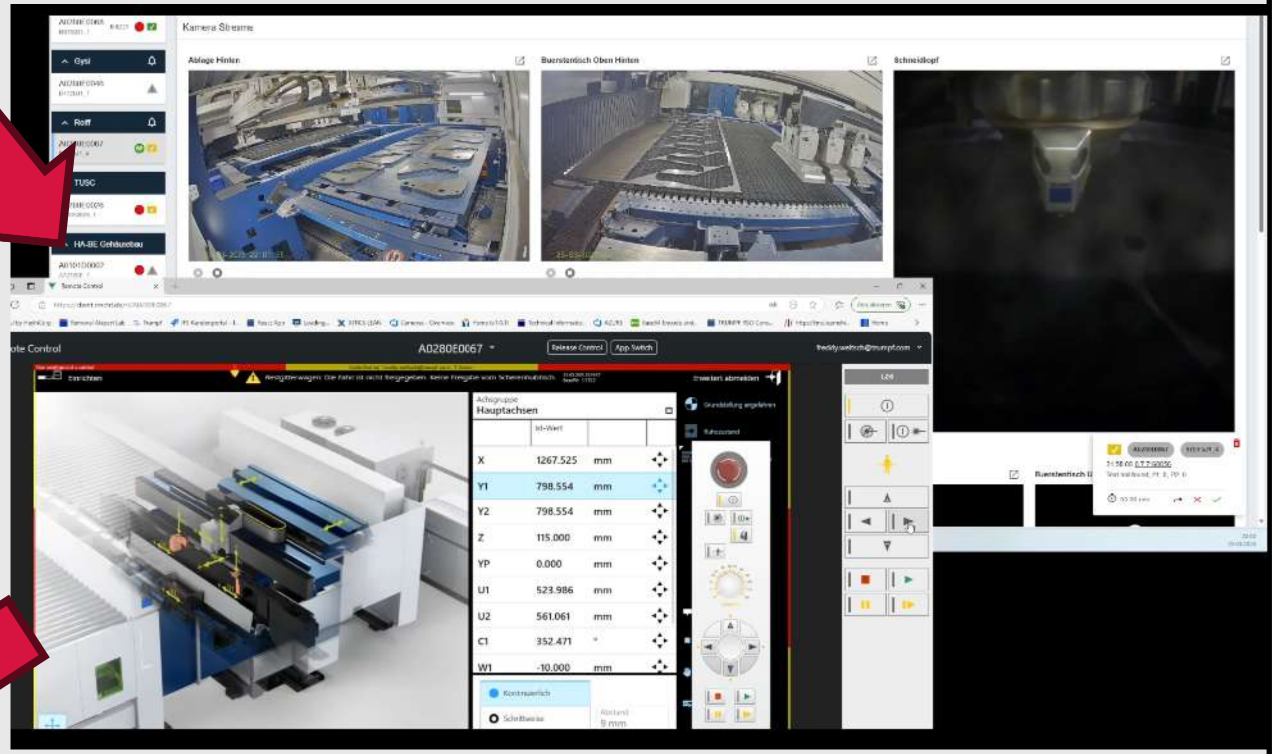
| source_timestamp | node_id | value |
|---|---|---|
| 2025-03-25T20:57:10.362+00:00 | 179 | {"ConditionI |
| 2025-03-25T20:57:12.381+00:00 | 179 | {"ConditionI |
| 2025-03-25T20:57:12.381+00:00 | 179 | {"ConditionI |
| 2025-03-25T20:57:15.367+00:00 | 179 | {"ConditionI |
| 2025-03-25T20:57:15.377+00:00 | 3039 | TRUE |
| 2025-03-25T20:57:15.377+00:00 | 3037 | 4 |
| 2025-03-25T20:58:08.721+00:00 | 53 | FALSE |
| 2025-03-25T20:58:08.721+00:00 | 54 | TRUE |
| 2025-03-25T20:58:08.721+00:00 | 59 | 1160 |
| 2025-03-25T20:58:08.731+00:00 | 3038 | 1 |
| 2025-03-25T20:58:08.731+00:00 | 3015 | FALSE |
| 2025-03-25T20:58:08.731+00:00 | 3034 | 0 |
| 2025-03-25T20:58:08.927+00:00 | 35.StoppedMalfunction | TRUE |
| 2025-03-25T20:58:08.927+00:00 | 35.Stopped | TRUE |
| 2025-03-25T20:58:08.927+00:00 | 35.Running | FALSE |

Detecting production faults

**Input multi-modal data:**

- Dynamic real-time videos
- Static knowledge graph
- Machine signals

Multimodal Large Language Model Analysis



Which camera allows for the fastest analysis of faults and derivation of solutions?

### Description:

Real machining data:

- 14 Camera angles, PLC data, OPC_UA error codes (100+, 1-2mins around failures, incl. resolution)
- Machining handbook and faults' descriptions .Pdf

> 300.000 tsd. Videos + machine data available for training

**Goal: Adapt autonomously to disturbances during the production**

-> Select best camera to view the error + propose resolution strategy based on best practices

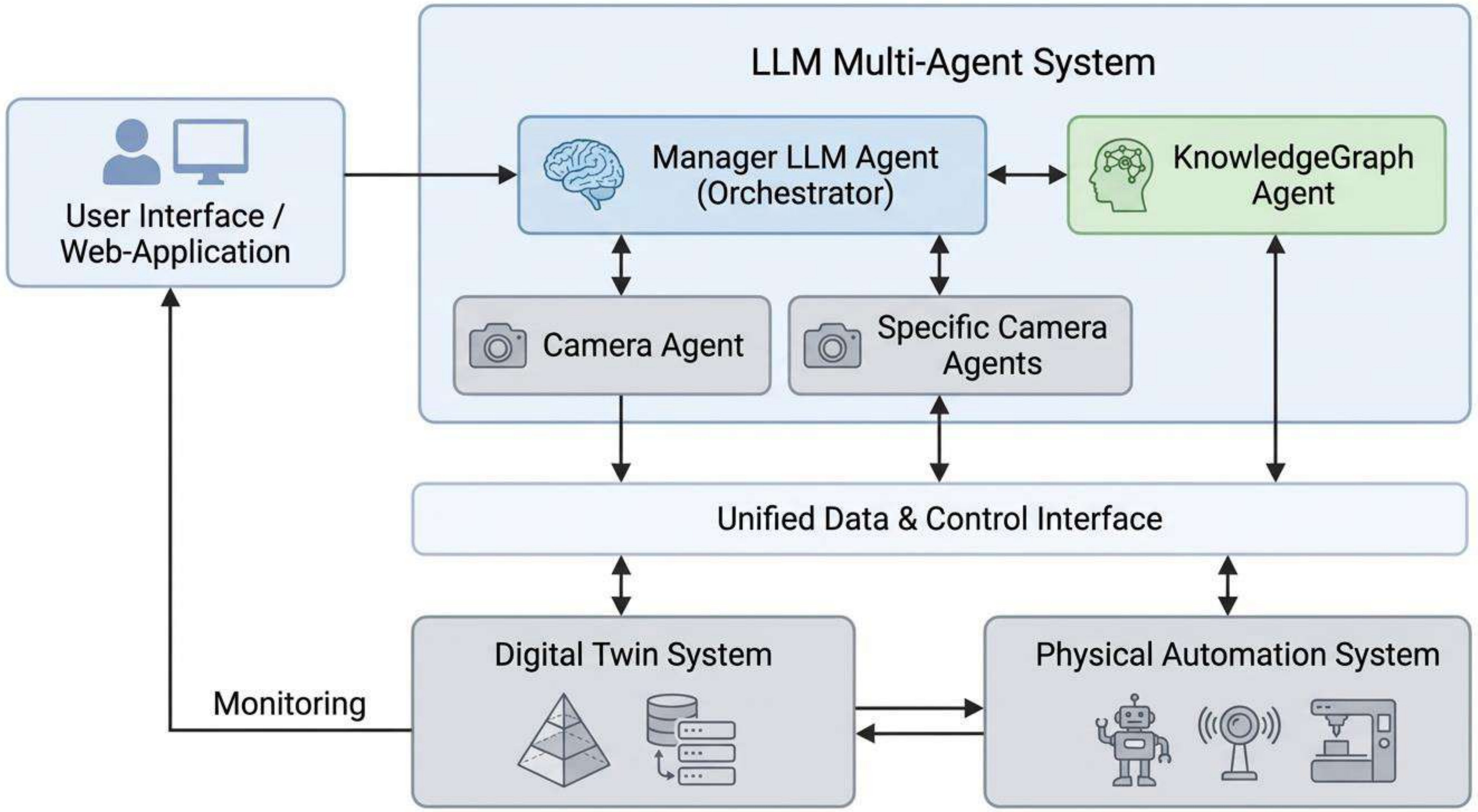
**Currently underway:**

- Static training: PDFs → graph-based knowledge for MLLM pretraining & querying
- Dynamic analysis: Video in OpenCV with VLM/MLLM (e.g., Qwen 2.5)

Expected outputs fast-response fault reports:

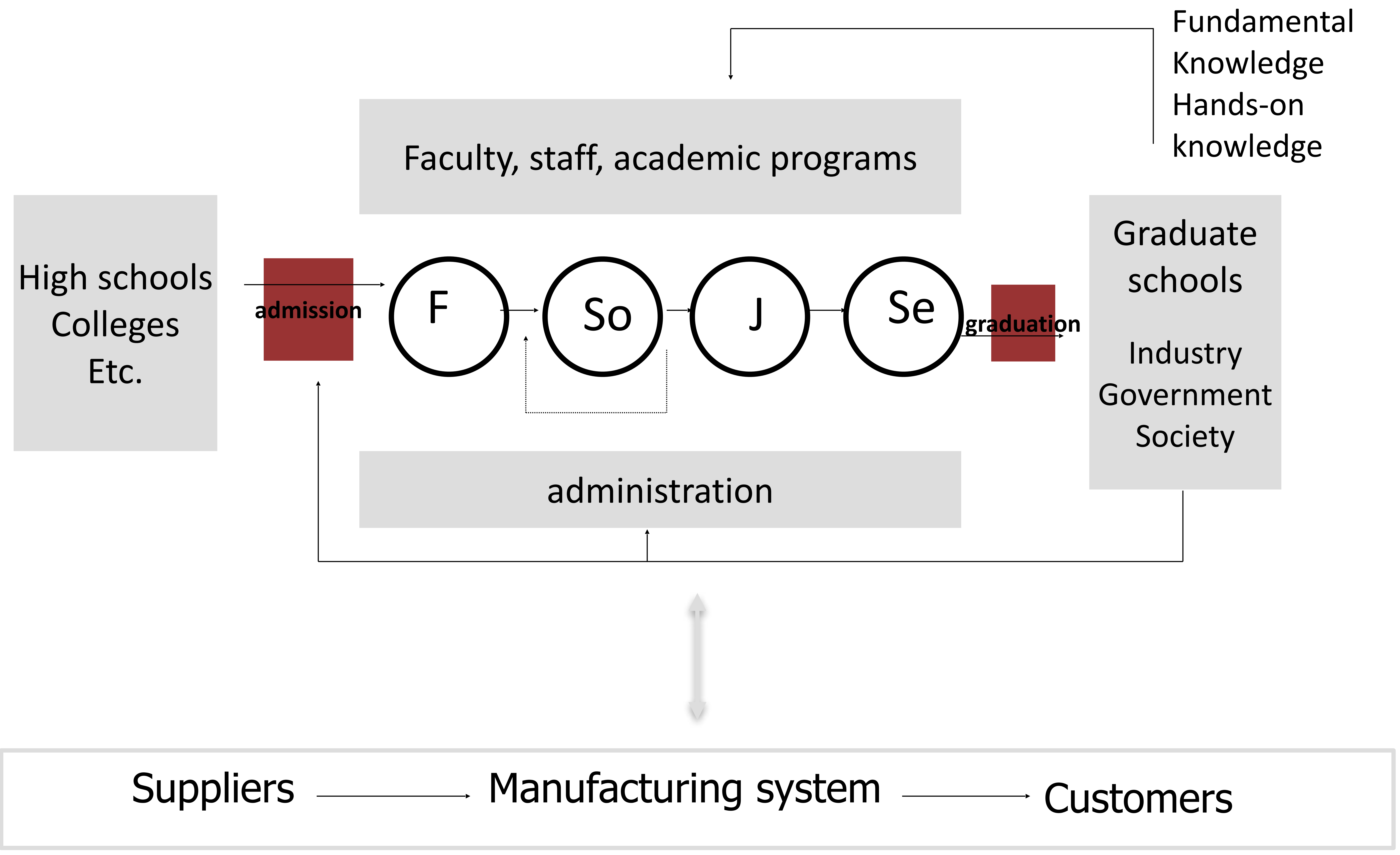
- Optimal camera view ranking
- Fault cause identification
- Recovery & mitigation strategies

# Multimodal LLM-VLM Multi Agent Solutions



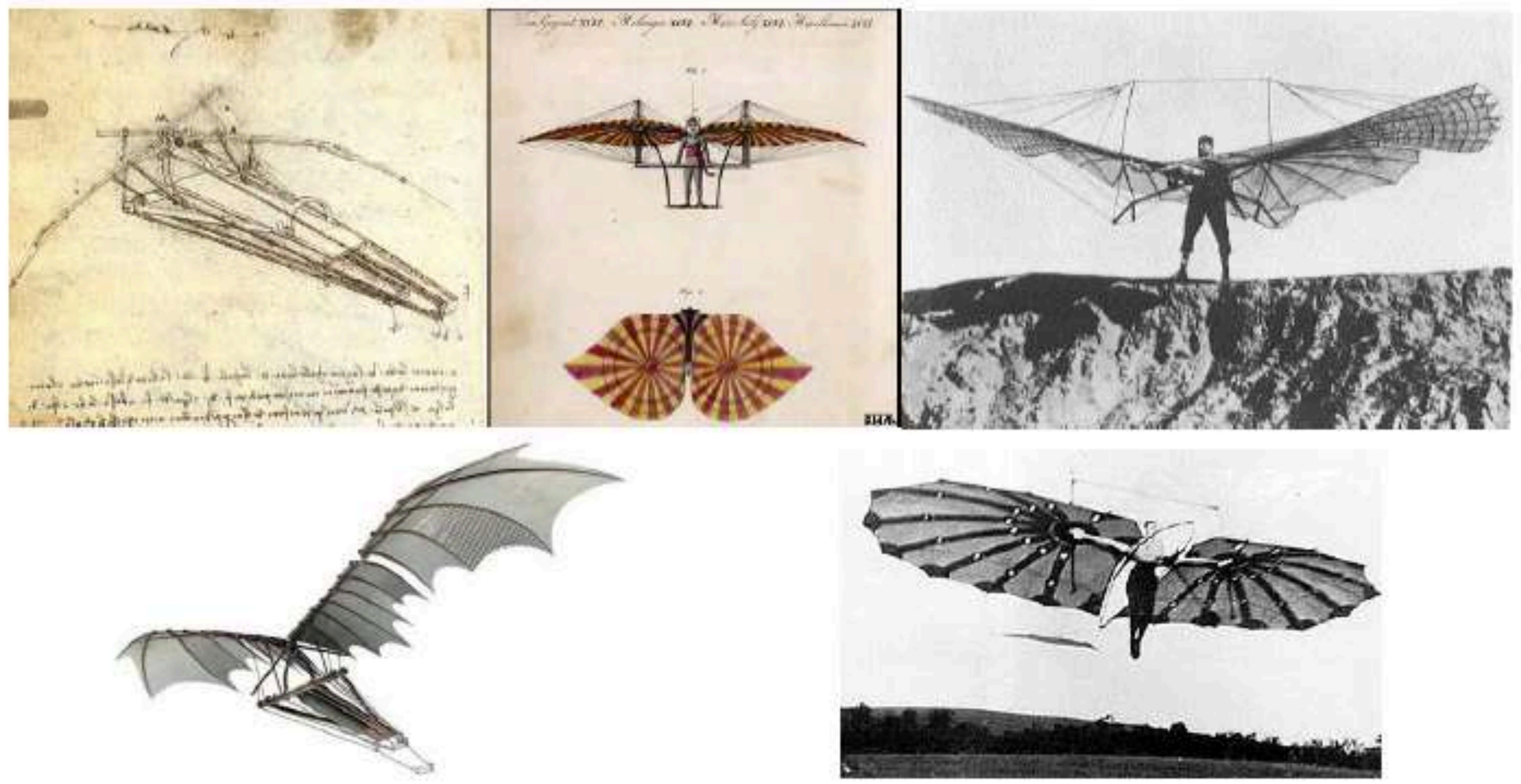

- Is this a functional view of physical view?
- How could we design AI Agents?

# A good University (in physical domain)



# A good University (in functional domain)

## Physical thinking



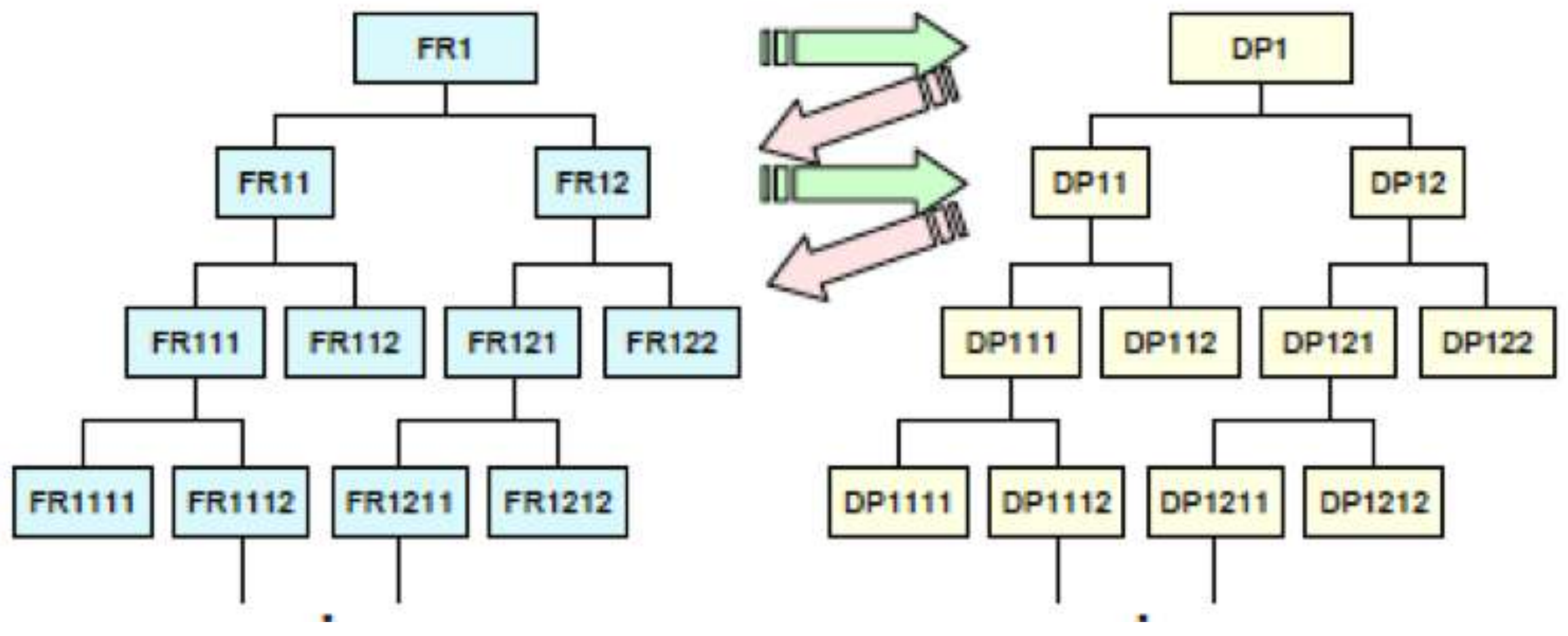
## Functional thinking

# Design of Agents

- Define functions (Functional Requirements)
- Separate “What’s (FRs)” from “How’s (DPs)”.
- Build "Tree of FRs" as high as possible.

  1. Buckets and "up & down"
  2. At the highest level, map $FR_1$ to $DP_1$ (or multiple DPs).
  3. No need to specify DPs at the highest level
  4. Then decompose $FR_1$ to $FR_{11}$ ..$FR_{1n}$.
  5. Map $FR_1$, .. $FR_{1n}$ to $DP_1$, ..$DP_{1n}$

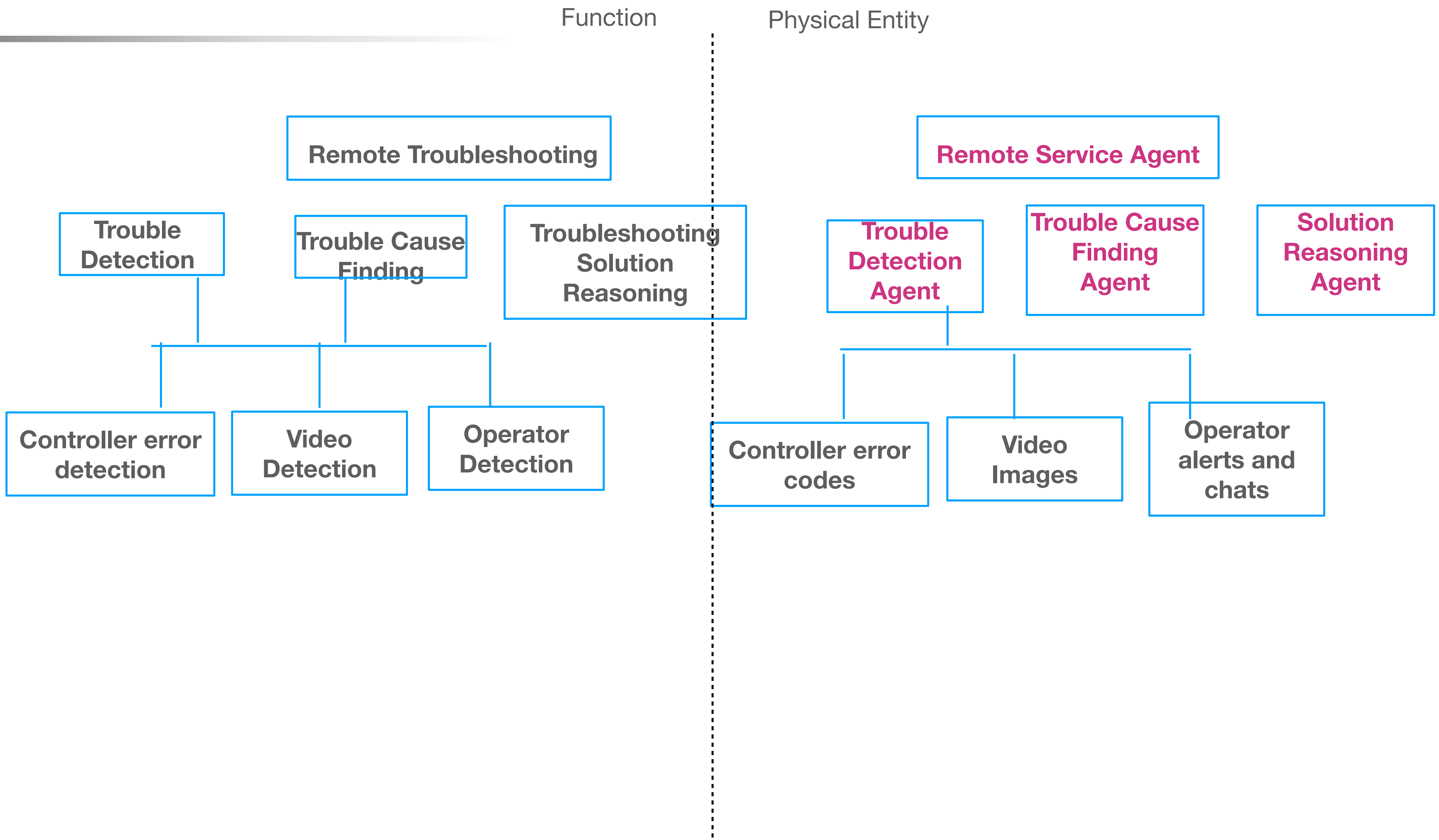
- Repeat




H. Akay, S. Kim "Design transcription: Deep learning-based design feature representation" CIRP Annals, 2020

# Design of Remote Troubleshooting Agents

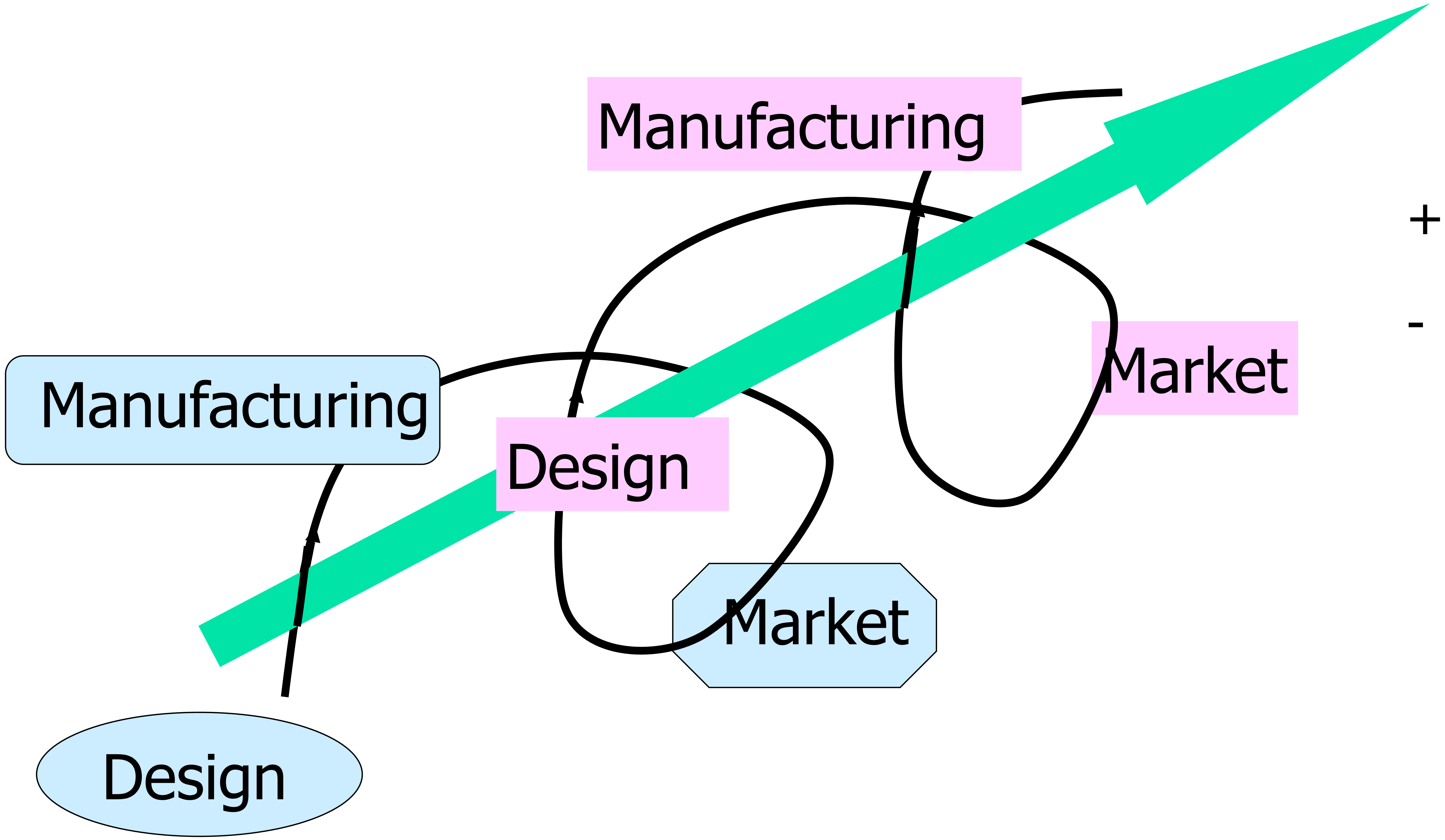
- Build "Tree of FRs" as high as possible.
  1. Buckets and "up & down"
  2. At the highest level, map FR1 to DP1 (or multiple DPs).
  3. No need to specify DPs at the highest level
  4. Then decompose FR1 to FR11 ..FR1n.
  5. Map FR1, .. FR1n to DP1, ..DP1n
- Repeat




H. Akay, S. Kim "Design transcription: Deep learning-based design feature representation" CIRP Annals, 2020

# Manufacturing Primer

# Manufacturing Helix




+ spiral: business

- spiral: charity

# What is Manufacturing

- Manufacturing vs Fabrication
- Transformation of materials and information into goods for the satisfaction of human needs.
- What is good manufacturing?
- Good restaurant?

- Quality
- Rate
- Cost
- Flexibility

⇨

- Profit, of course!
- Productivity
- Competitiveness

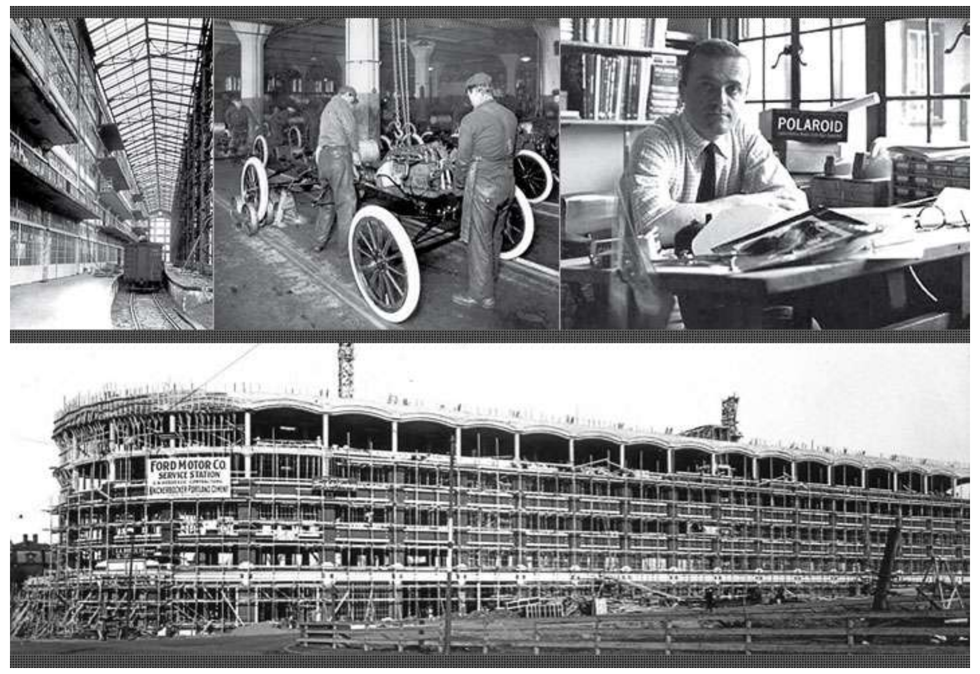
# Industrial Leaderboard Reshuffling after Key Technological Disruptions

- SONY (Analog) to Samsung (Digital)
- IBM (Main frame) to MS/Apple (Personal Computing)
- GM/Ford to Toyota to Tesla/Hyundai/who? (Electric Vehicle)
- Low-cost-labor Manufacturing v.s. Smart Manufacturing (AI?)

# Big Picture: AI for Manufacturing

- How can AI achieve for us what is currently beyond the capability of others?

at 640 Memorial Dr.



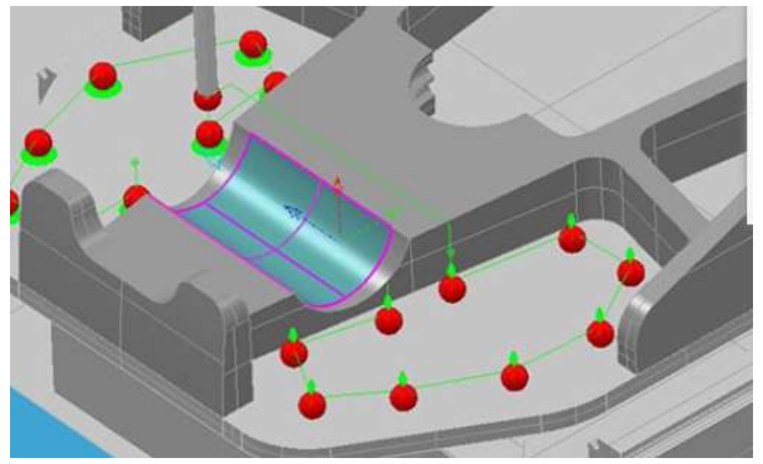

Ford Motor Factory ~1920's., Cambridgeport

# Evolution of Manufacturing System Technologies

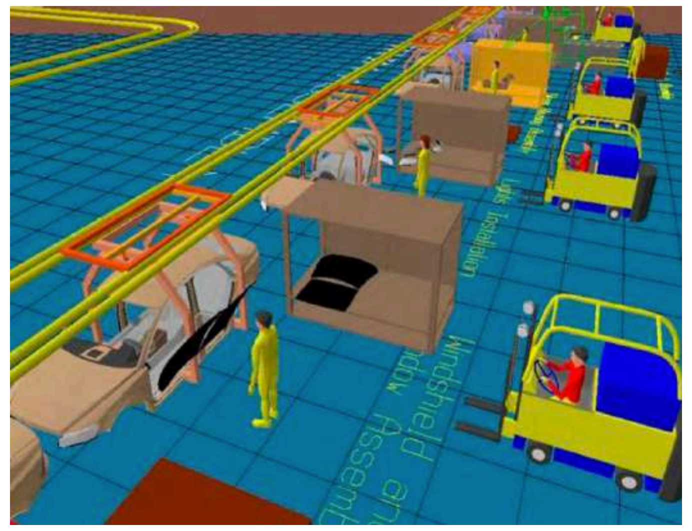
Opportunities

Ford Model T moving assembly, 1913


MIT NC Machine, 1952



CAD/CAM, 1970/80



Computer Integrated Manufacturing. 1980/90

**TPS (Toyota Production System)**
**JIT (Just in Time)**
**LEAN Manufacturing System**

**AI for Manufacturing**

**Smart Manufacturing**

1990's

**Intelligent Manufacturing Systems**

2000's

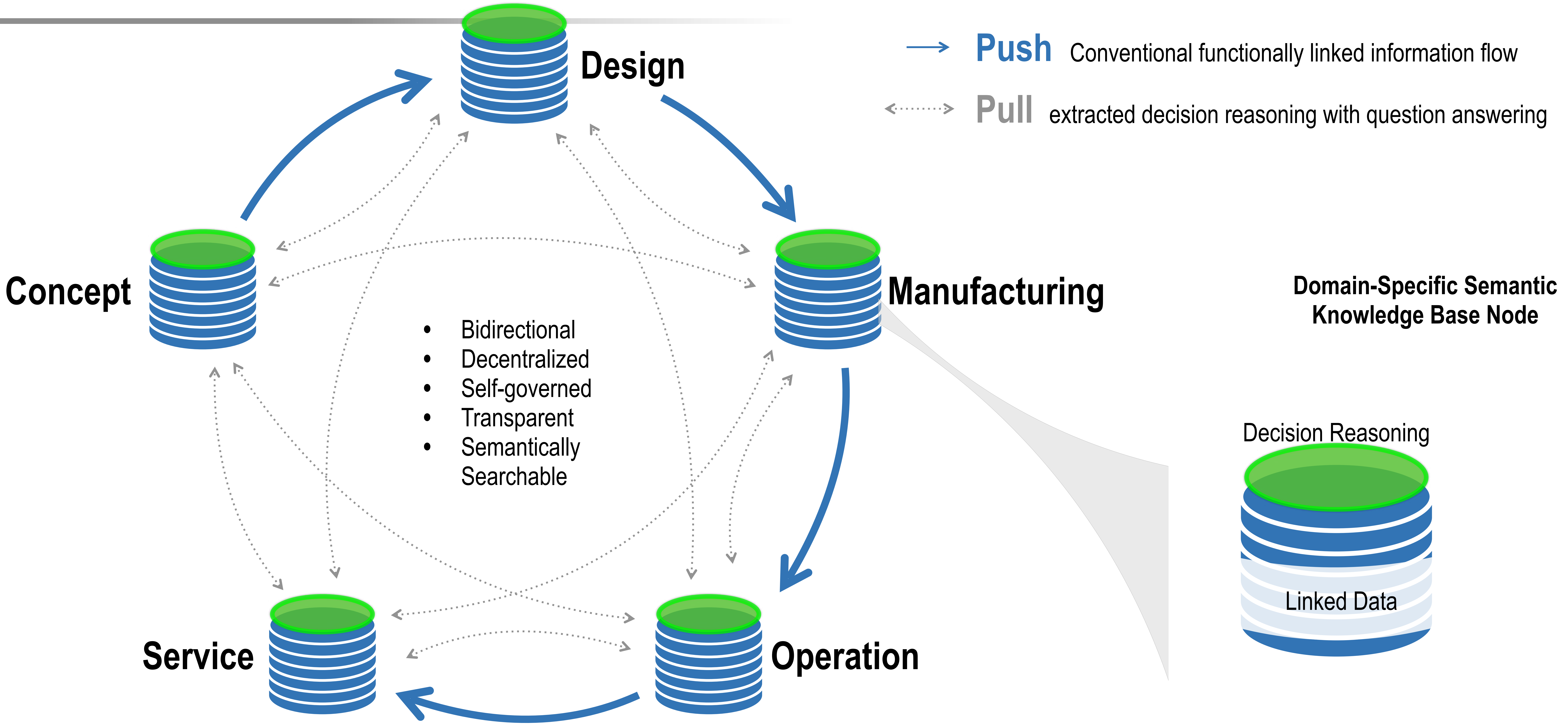
**Industry 4.0**
**Cyber Physical Systems**
**Digital Twins**

2010's

**Future Manufacturing?**

2030-

# Smart Manufacturing needs to have knowledge threading among decision makers.




- Bidirectional
- Decentralized
- Self-governed
- Transparent
- Semantically Searchable

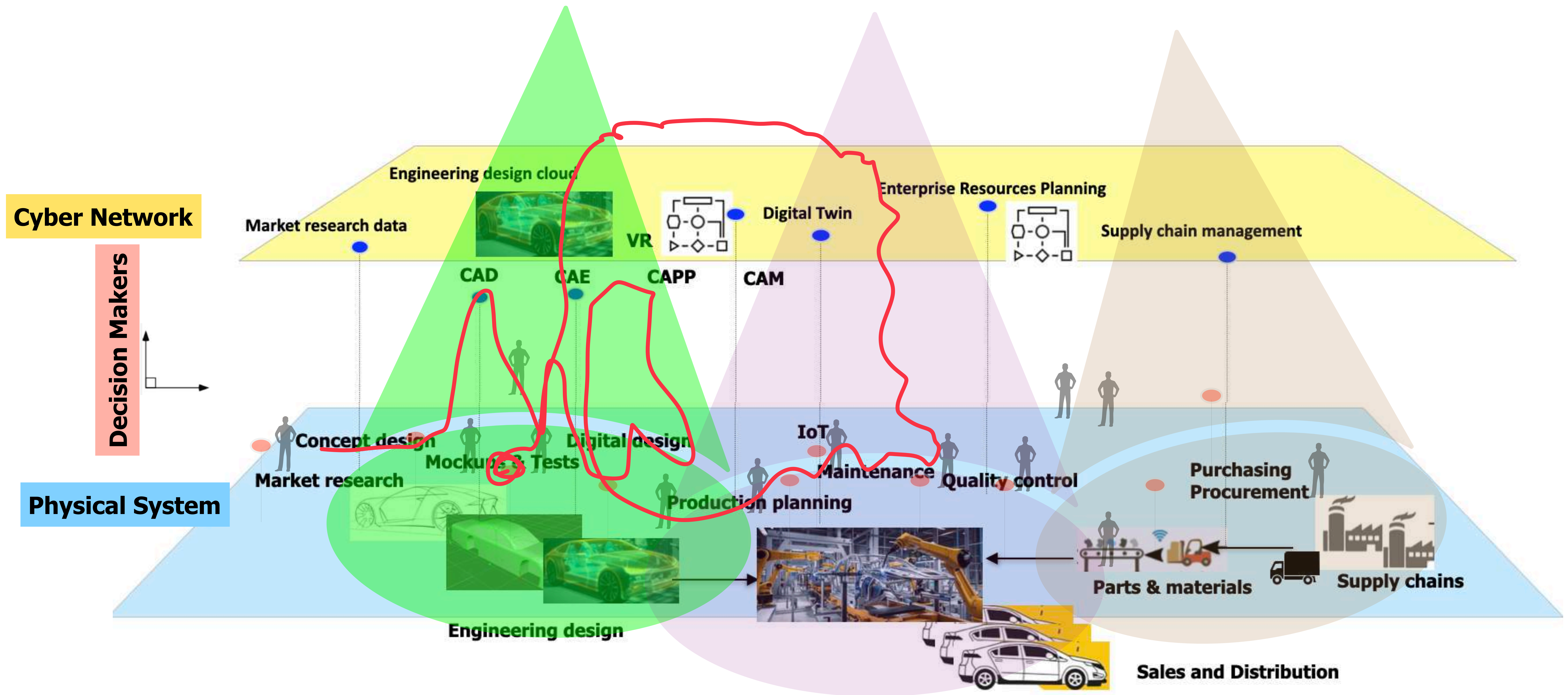


Haluk Akay, Sang Hyun Lee, Sang-Gook Kim, CIRP Annals
Volume 72, Issue 1, 2023, Pages 401-404

Sang-Gook Kim, MIT AIM Group

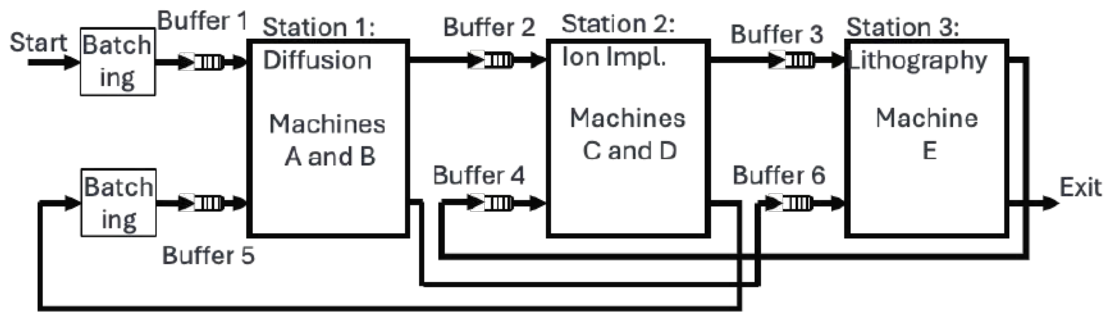
# Fully Digitalized MFG Systems are not necessarily Truly Smart Systems: Why?

**Planar Cyber-Physical Systems + Vertical Siloed Decision Makers" —> Trouble for Full Integration.**



# Foundational Platform of Manufacturing Agentic AI

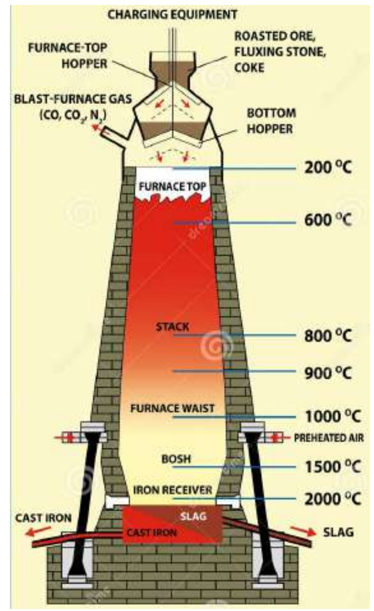
INM project, Paul Liang and Sang-Gook Kim

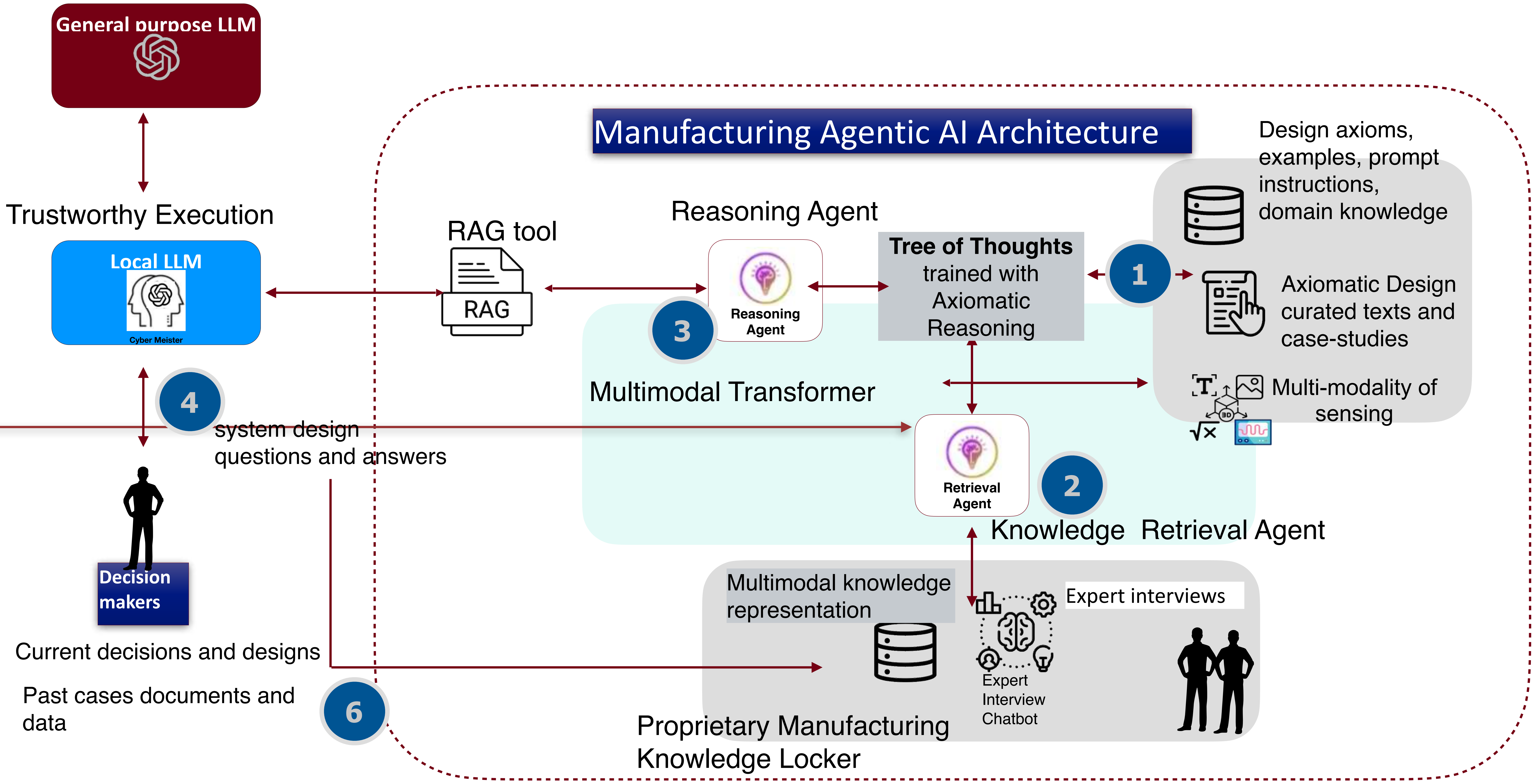



Intel MiniFab

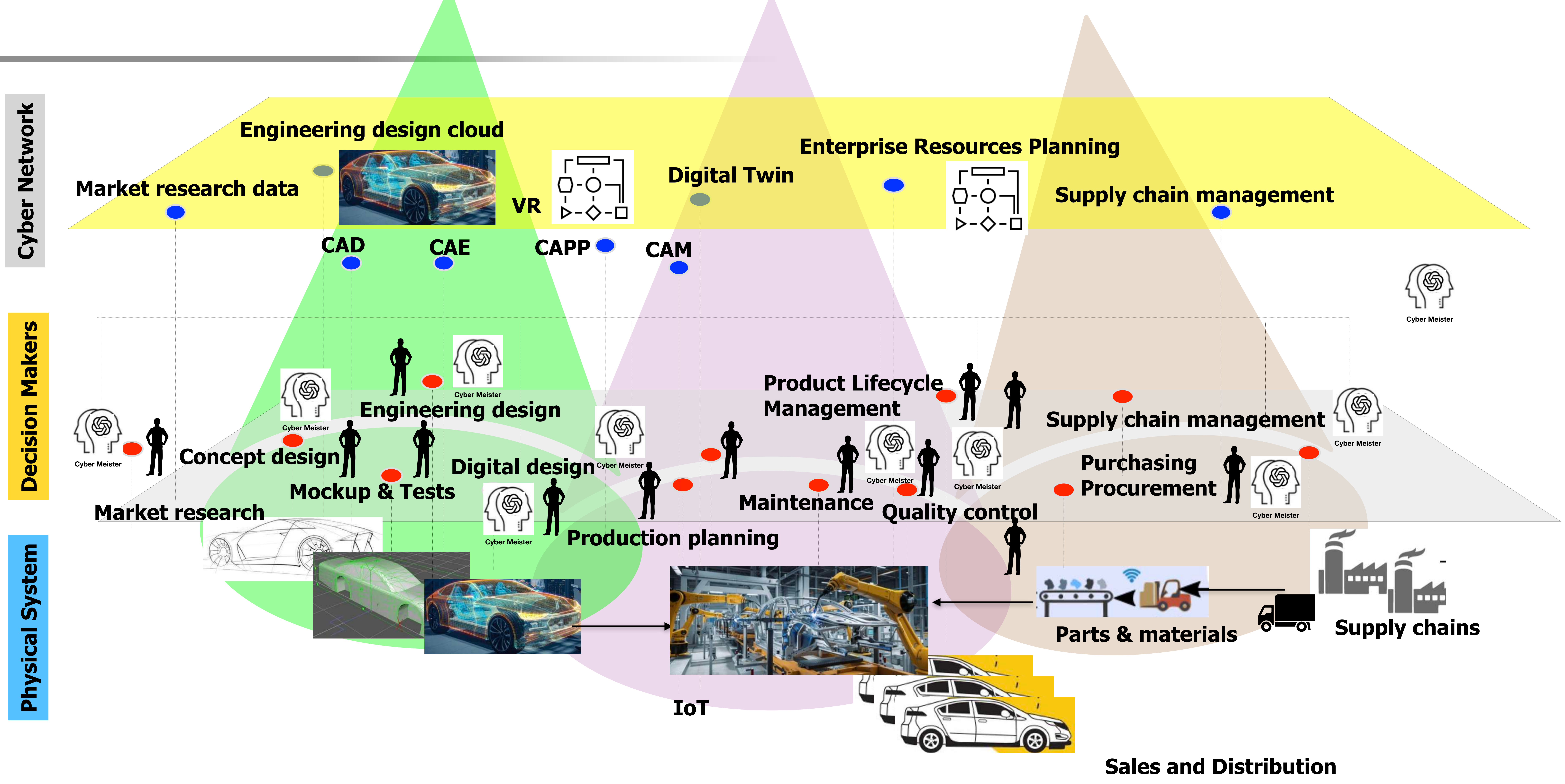
- INM interest group systems
- Trumpf Machines
- 3D Metal Additive Manufactring
- Grid-level storage
- Blast Furnace

…





# Future of Manufacturing: with Decision Agents



# Digital Transformation of Design and Manufacturing

Digitalized Operations (current) to Digital Enterprise (future)

- Smart Manufacturing, Prof. Hatvany, CIRP Conference 1985
- Industry 4.0, 2000’s
- Digitalization, Digital Twin, Cyber Physical Systems, etc.,, 2010’s
- Digital Transformation (Manufacturing and Management)